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Major Project: Report

on

DESIGN OPTIMIZATION OF ELECTRIC 3-WHEELER CABIN TO MINIMIZE VIBRATIONS AND IMPROVE DRIVER SAFETY MPD491P

Submitted by
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of

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in
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DEPARTMENT OF MECHANICAL ENGINEERING

2023-24

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CERTIFICATE

Certified that the project work titled "**Design optimization of electric 3-wheeler cabin to minimize vibrations and improve driver safety"**, carried out by **ADITYA KASHYAP, USN: 1RV22MPD02**, a bonafide student, submitted in partial fulfilment for the award of **Master of Technology** in **Product Design and Manufacturing** of **RV College of Engineering®, Bengaluru, affiliated to Visvesvaraya Technological University, Belagavi,** during the year **2023-24.** It is certified that all corrections/suggestions indicated for internal assessment have been incorporated in the report deposited in the departmental library. The project report has been approved as it satisfies the academic requirement in respect of project work prescribed for the said degree.

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DECLARATION

I, Aditya Kashyap, student of fourth semester M. Tech in **Product Design and Manufacturing Department of Mechanical Engineering,** RV College of Engineering®, Bengaluru, declare that the Major Project with title **"Design optimization of electric 3-wheeler cabin to minimize vibrations and improve driver safety",** has been carried out by me. It has been submitted in partial fulfilment for the award of degree in **Master of Technology** in **Product Design and Manufacturing** of RV College of Engineering®, Bengaluru, affiliated to Visvesvaraya Technological University, Belagavi, during the academic year **2023-24**. The matter embodied in this report has not been submitted to any other university or institution for the award of any other degree or diploma.

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GREAVES ELECTRIC MOBILITY

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This is to certify that **Mr. Aditya Kashyap**, bearing the USN: 1RV22MPD02, pursuing M. Tech in Product Design and Manufacturing from R V College of Engineering, Bengaluru, has successfully completed his project on **"Design Optimization of electric cargo three-wheeler cabin to minimize vibrations and improve driver safety"** at Greaves Electric Mobility, Yelachenahalli, Bengaluru from **8th February 2024 to 8th August 2024**.

Supriya Thankappan,

Head HR

Greaves Electric Mobility Private Limited

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ABSTRACT

The Global Three-Wheeler Market, valued at USD 13.93 billion in 2024, is expected to grow at a CAGR of 10.98% to reach USD 39.48 billion by 2033.Unreliable. Lack of accurate road vibrations data of electric three wheelers result in R&D teams designing vehicles with unsafe vibration transmissibility on driver body. Previous work carried out RLDA on cars, trucks, EV two wheelers but no attempt has been made specifically to collect RLDA data of electric three wheelers and optimization of design. The objective of the research work was to collect frequency responses of electric 3-wheeler and carry out RLDA for optimization of cabin design to keep vibrations on driver body below unsafe limits.

First, benchmarking of L5 category vehicles was done at Greaves electric mobility, for finalizing a proto vehicle. Specifications of vehicle were used to prepare a CAD model in SEIMENS NX. The CAD model was imported into ANSYS workbench for modal analysis with optimized mesh with Jacobian index 0.63 of SOLID 187 element maximum size 8 mm, with free-free boundary condition. Experimental modal analysis was carried out for chassis using accelerometers and impact hammer and FFT analysis in DEWESoft software. First two significant bending modes in vehicles Z-axis and Y-axis. with natural frequencies 22 Hz and 66 Hz. first two significant torsional modes along Z-axis and Y-axis at 88 Hz and 101 Hz. And one torsional and bending mode at 125 Hz. 5 to 8 % difference in FEA results ad experimental values is observed. Further, Using DEWSoft software and instruments, Road Load Data Acquisition (RLDA) and Human vibrations collected for a male driver in 35, 45 and 55 kmph with road roughness index range 1.2 m/km (smooth road) to 5 m/km (bad road) and loads varied from 150 kg to 390 kg based on the DOE using Taguchi L27 array to perform ANOVA and RSM optimization.

VDV values isolated at steering column and seat ranged from 5 to 16 due to 42 % transmissibility of vibration from motor mountings, and 49% from steering column and chassis. ANOVA results after obtaining responses showed VDV is significantly affected by road roughness ($F = 15679.65$, $p < 0.001$), vehicle speed ($F = 1753.79$, $p < 0.001$), and loading ($F = 105.74$, $p < 0.001$). regression analysis show, baseline value of VDV is -0.8462 when all variables are at 0. The R^2 value indicates 89.9% variability in VDV by the independent variables (F-statistic: 5532) indicating unit increase in road roughness raises VDV by 1.1280.5 $m/s^{1.75}$, vehicle speed increases it by 0.0737 $m/s^{1.75}$, and loading increases it by 0.0013 $m/s^{1.75}$. optimization through RSM indicated the need to decrease spring stiffness and increase mass of steering column. Hence, these design inputs were used to design a U-fork steering and suspension with modified spring and a seat mount.

New U-fork steering column with modified suspension spring and L-shape seat mount was designed and implemented in vehicle followed by testing showed maximum VDV recorded as 11.59 $m/s^{1.75}$. Future scope of work can use adaptive damping shock absorber for varying road RIs and loads of vehicle.

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ABBREVIATIONS

Symbol	Description
CAGR	Compound Annual Growth rate
FEA	Finite Element Analysis
FFT	Fast Fourier Transform
EMA	Experimental Modal Analysis
RLDA	Road Load Data Acquisition
WBV	Whole Body vibration
HAV	Hand Arm Vibration
ANOVA	Analysis of Variance
RSM	Response Surface Methodology
DOE	Design of Experiments
VDV	Vibration Dose Value
IRI	International Roughness Index

CHAPTER 1

INTRODUCTION

1.1 Market Research

Electric three-wheeler manufacturers have to comply with ISO 2631 standards of mechanical vibration and shock for whole body vibrations and ISO 5349 for Hand-arm vibrations. In the ongoing research and development electric three wheelers are facing of delayed vehicle certifications from government agencies such as ARAI, if PSD values cross more than 0.01 g^2/Hz from random road vibrations. Electric three-wheeler market is booming across the world at a rapid rate. The Specialty Equipment Market Association (SEMA) report projects the global three-wheeler steering and suspension parts market to grow at a 6.5% Compound Annual Growth Rate (CAGR) from 2024 to 2030. Valued at $1.2 billion in 2024, it is expected to reach $2.1 billion by 2030. All the parts need to handle vibration levels and should be designed such that vehicle and road induced vibrations are not entirely passed on to the driver and passengers. Figure 1.1 shows the global market players who make steering columns, suspensions assemblies and driver seat fittings for three-wheelers. [1]

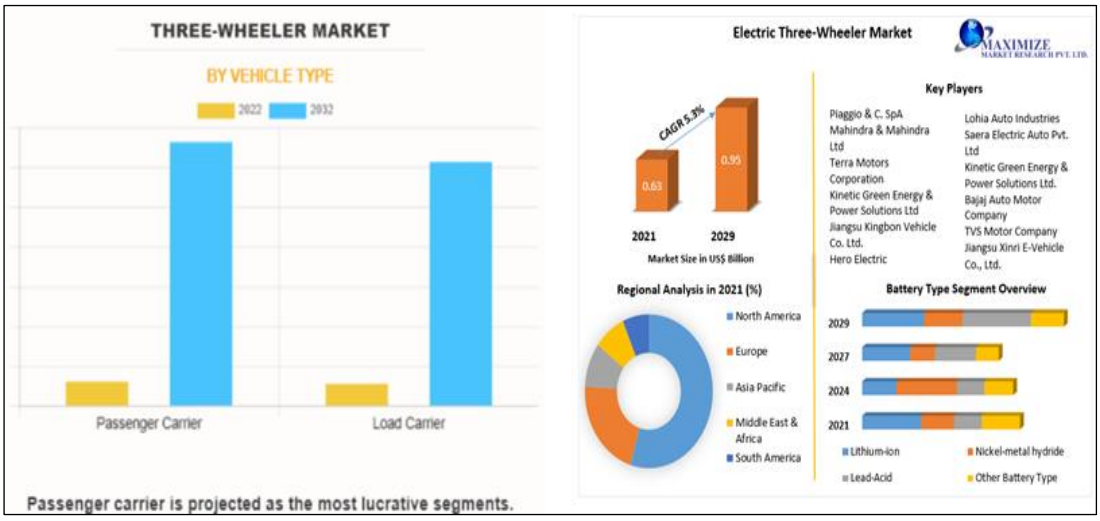


Figure 1.1: SEMA market statistics- 2024 of 3-wheeler parts

1.2. Literature survey

1.2.1. Driver discomfort surveys

The dynamic simulation of commercial transport truck in CARSIM found that at ride speeds of 40 km/h and at manhole depths of 7.5 cm and above driver body experiences 0.5 m/s^2 of vibration.[1]. Effects of electro-magnetic active seat suspension were evaluated that reduced exposure of a long-haul truck driver to whole body vibration (WBV) on low back pain. WBV exposures decreased post seat installation for active seat suspension but not for passive seat [2]. Detailed comparison and discussion of methods were defined in the European Directive and the is used to form a methodology for the management and organization of preventive measures to vibrations exposure [3]. FFT analysis is a must for studying vibration characteristics of a system. Modal test of an auto chassis at 63 points, with the help of Zonic A&D fifth-generation modal analysis software with Mass and stiffness optimization to simulate a fully loaded vehicle, were correlated with the resonant peaks experienced in the auto running at 30 km/h on typical bad roads. [4]. Suspension plays an important role in damping road induced vibrations. The vertical acceleration for modified spring and old spring were measured as 10 m/s^2 indicating no drastic change in ride comfort and handling characteristics of the vehicle [5].

The frequency responses of the model of 4 DOF vibrating system were studied. Rolling acceleration mean value among the resonance band in the frequency response curve was reduced by 60%. [6]. Vibration measurements were carried out to obtain experimental whole-body vibrations of 10 healthy male subjects and compared under different road conditions [7]. As per ISO 2631-1 standards whole-body vibration exposure measurements were recorded at the operator–seat interface with a tri-axial seat pad accelerometer using a vibration analyzer. Transport equipment operators were exposed to higher vibrations compared to non-transport equipment operators [8]. Prediction model of vibration transmissibility was developed from input design parameters using the response surface method. The vibration transmissibility ratio of the optimized seat system is reduced by about 10% [9]. Vibration levels of a common 200 cc cross motorcycle cross 30 Hz and a correct positioning of the saddle can minimize the whole-body vibrations [10].

1.2.2. Modal analysis

The final optimized chassis achieved a significant reduction of 42% in maximum deformation and a shift in the lowest modal frequency from 11.691 Hz to 57.318 Hz. Torsional rigidity and crashworthiness were also improved, resulting in a more robust and reliable chassis design. [11] In the modal analysis, modes 1 to 10 is found and the natural frequencies lie in the range 15.62 Hz to 90 Hz and a low mass participation factor of 30% due to which resonance is unlikely to occur within the chassis, thereby enhancing its stability and reducing the risk of resonant vibrations [12]. Modal FEA in 3 configurations of direct drive in-wheel motor Ev with reduction and transmission gear, reveals 24 Hz additional mode and 4 Hz initial mode that might cause resonance in ride condition if not addressed. [13]

1.2.3. FEA structural

CAD modelling in SOLIDWORKS and structural FEA in ANSYS of a rear-wheel-drive FSAE vehicle chassis for an electric car with 80 kmph top speed is done and followed by manufacturing to integrate a single seating, motor and battery. [14]

Inertial relief technique for FEA EV 3-wheeler chassis in ANSYS with external, rotational and torsional loads for a Hex meshed model of 5 mm size, shows modified single type chassis has more strength, light in weight and easy to fabricate compared to existing double type chassis with FoS calculated to be 1.874. [15]

1.2.4. Human Vibrations

The vibration characteristics and occupant comfort in a three-wheeled Indian auto-rickshaw using data was collected from a 3-axis accelerometer. The experiment involved four driving scenarios, including idle, constant speed cruising, acceleration, and deceleration. The presence of low-frequency vibrations (0.5 Hz) leading to motion sickness and higher frequency modes (10 Hz, 25 Hz) associated with discomfort were found and Simulink was used to model vertical vibrations, leading to the conclusion that softer suspension improves comfort, particularly at 50 km/h, where resonance frequencies increase discomfort. [16]

Ride comfort analysis of 19 DOF heavy truck model is evaluated on long span bridge under different loads and traffic simulations which revealed overall threshold vibration value was below 0.315 m/s^2. [17]

Operational Modal Analysis and simulations of In-wheel-motor EV for different slip ratios for motor coupling is done and compared. PSD results of ride test is used to propose advanced control design for primary (1-3Hz) and secondary (3-10 Hz) vibrations to improve motor design. [18]

1.2.5. RLDA

Simulations on CARSIM for different ride conditions of truck on gradient with 0.35 road friction coefficient showed PSD values between 10^{-3} g^2/Hz and $10^{-1}g^2$/Hz which is acceptable for heavy duty vehicles as per ISO 8608.

Modelling of artificial road profiles for 200 km range for different conditions showed PSD values obtained through MATLAB code. Road roughness was 12 % lower than international roughness index. [19]

The e-rickshaw testing on a chassis dynamometer using the EVIDC cycle, with energy consumption around 20.87 Wh/km (standard deviation 3.57) and validation with EVIDC cycle of average speed 13.79 km/h, a duration of 1125 seconds, and 4.31 km confirms consistency between on-road and dynamometer energy consumption simulations. [20]

1.2.6. Optimization

Nagarkar et al. developed a nonlinear quarter car suspension–seat–driver model with 4 DOF, incorporating quadratic tire stiffness and cubic suspension spring stiffness for optimal design. The study focused on optimizing ride comfort and health criteria, such as Vibration Dose Value (VDV) at the head, frequency-weighted RMS head acceleration, and crest factor, along with stability criteria like suspension space deflection and dynamic tire force. ISO 2631-1 standards were applied to assess these objectives. Optimization was achieved using NSGA-II and MOPSO-CD algorithms, with the latter offering greater computational efficiency. Simulations for a quarter car on a Class C road at 80 km/h demonstrated that the optimized design improved ride comfort and health metrics over classical designs. [21-22] Vibration attenuation through the suspension and seat will not only provide riding comfort but also reduce the risk of low back pain due to driving. One of the early studies on the biomechanics of seated drivers subject to vibration was realized and optimization algorithms were used [23]

Simulation computations focusing on the analysis of overturning stability revealed lighter vehicle exhibits better dynamics and driving performance in curves and, the modified vehicle, with higher total weight, is safer in terms of vertical dynamics, reducing the risk of overturning showing trade-off between agility in handling and stability under vertical loads. [24] Humans exposed to sinusoidal vibration and impact and a detailed experimental work on translational seat vibration was performed to determine the effects of level, frequency and direction of the seat vibration [25].

Dynamic response of a seated subject was investigated in various aspects, for example, the effect of various cushions he effect of vibration frequency and posture and the effect of backrest established a seated human body model to design an optimal seat suspension for isolation of the vertical whole body vibration based on the simulated subjective response [26]. The objective function combining the head acceleration, its crest factor, the suspension deflection, and the tire deflection are used to provide not only the stability of the car but also the comfort of the driver [27]

On the other end of the spectrum, the effect of spinal forces due to whole body vibration and sitting biomechanics were considered in this study [28]

1.3. Research Gap

Researches [2-25] have surveyed cabin of trucks, cars, mining equipment in HTV ,LTV for driver discomfort with VDVs more than 16m/s$^{1.75,}$ and done modal analysis using piezo-sensors on cargo vehicles with natural frequencies from 20 Hz to 600 Hz up to 10 significant modes, dynamic simulations in CARSIM and MATLAB with quarter-car models of 4, 6, 9 DOFs, RLDA on roads of roughness index 1.2 m/km to 6m/km resulting in PSDs from 0.001 g^2/Hz to 0.02 g^2/Hz while rest carried optimization using L 27 array with RSM and GA after ANOVA and Regression analysis with R^2 varying from 70% to 96% fit for factors such as vehicle speed, road roughness and loads influencing VDVs. But so far, no attempt is made to modify design of a component contributing to vibration specifically in electric 3-wheelers and reduce vibration below 8 g^2/Hz VDV on drivers. Therefore, the research involves investigation of vibration response of vehicles through Modal analysis and RLDA and driver vibrations using DEWESoft software and design a steering column and seat mount to reduce vibration dose values below 8 g^2/Hz.

1.1.8. Project Objectives

The project aims to achieve the following objectives:

1. To find out resonant frequencies of the proto vehicle through modal analysis.

2. To carry out RLDA and whole-body vibrations test at critical ride conditions and loads.

3.To perform ANOVA and regression analysis to find out individual contributions of ride conditions on driver body vibrations.

4.To perform response surface optimization and implement modifications in vehicle to reduce VDVs on driver.

1.1.9. Organization of the report

Chapter 1 INTRODUCTION: Introduction to 3-wheeler market from SEMA 2024 market statistics of three-wheeler parts and compound annual growth rate from 2024 to 2030.

Chapter 2 CONCEPT & THEORY: Fundamentals of Modal analysis, RLDA and while body vibrations

Chapter 3 OBJECTIVE AND METHODOLOGY: Defines the objectives of the project work and the methodology adopted to attain the defined objectives.

Chapter 4 SIMULATION SETUP: Discussion of ANSYS workbench modal analysis pre-processing, mesh convergence and solutions.

Chapter 5 EXPERIMENTAL INVESTIGATION: RLDA data collection is elaborated as per design of experiments using Taguchi method using L27 orthogonal array.

Chapter 6 DESIGN INPUTS: Inputs from optimized results to make modification in mass and stiffness of steering column and material selection.

Chapter 7 DESIGN & MODELLING: Design and analysis of new steering column for reducing vibrations doe values below 8 $m/s^{1.75}$

Chapter 8 RESULTS & DISCUSSION: Results obtained from modal analysis, RLDA and optimization.

Chapter 9 CONCLUSION AND FUTURE SCOPE: Presents conclusions of the electric 3-wheeler vibration dynamics and driver body vibrations and propose future scope of work.

CHAPTER 2

THEORY AND CONCEPTS

2.1. Vibrational dynamics

2.1.1. Modal analysis

is the study of dynamic characteristics of a structure in the frequency domain. The three main components of a mode are explained below:

Natural frequency: It is the frequency at which a system naturally oscillates in the absence of external forces. For a simple mass-spring system, natural frequency is given in equation (1) as

$$f = \frac{1}{2\pi}\sqrt{\frac{k}{m}}$$ **……..[equation 2.1]**

Damping Ratio (ζ): A dimensionless measure of how oscillations decay in a system, shown in equation (2),

$$\zeta = \frac{c}{2\sqrt{k * m}}$$ **……..[equation 2.2]**

ζ = damping ratio (dimensionless)

c = damping coefficient (Ns/m)

k = spring stiffness (N/m)

m = mass (kg)

Vibration Modes and their effects in vehicle

1. Bending mode [mode 1]: it involves flexural deformation of vehicle chassis and cabin lead to fatigue failures of critical joints at ride conditions from motor mounting loads and increases vibration transmissibility on driver.

2. **Torsional mode [mode 2]:** It involves twisting or torsional deformation of chassis and cabin structure. Excessive torsional loads lead to resonance effects in ride condition.

3. **Axial modes [mode 3]:** It involves longitudinal deformation long chassis and cabin length. They affect overall dynamic behavior of vehicle and vibration transmissibility.

4. Flexural-torsional mode [mode 4]: Both bending and torsional deformations. They can be particularly critical in steering column handling and chassis structural integrity. By understanding these modes, design of steering, suspension and seat mounts can be effectively done for safety of driver during ride.

2.1.2. Experimental Method

The structure is mechanically excited and response is obtained through sensors mounted on all the critical locations. Since the output is initially in time domain, the Time-domain to Frequency-domain conversion is done through Fourier analysis, any signal can be expressed as a combination of different sine and cosine waves with various frequencies and amplitudes. Recombining these separated components closely approximates the original signal. The fast Fourier transform (FFT) is an algorithm that calculates the discrete Fourier transform (DFT) of a sequence (Fourier analysis). Spectrogram representation - FFT over time allows us to visualize the changes in the frequency spectra over time, as obtained using FFT algorithms. To generate a spectrogram, the input time-history signal must be divided into smaller "frames," each of which is then converted to frequency spectra using FFT algorithms. Visualizing the frequency domain and its advantages: Resonance study is crucial in structural dynamics to prevent catastrophic damage from resonant excitations.

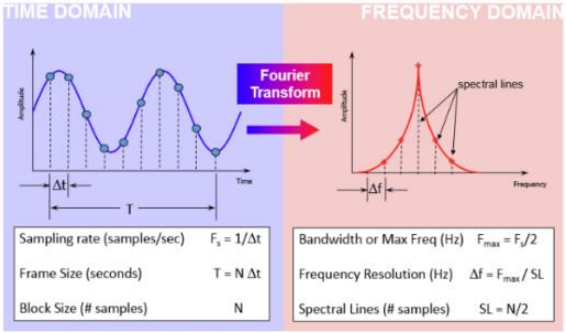


Figure 1.2 FFT analysis to convert data from time domain to frequency domain

Frequency domain analysis is particularly valuable in CAE vibration/fatigue analyses because it offers a faster and more efficient way to determine system responses compared to calculating time-series responses event by event. The frequency response function (FRF) is calculated at each measurement location and the modal parameters of the structure are derived. In equation() , FRF 'H(ω)' is defined as the ratio of the output response of a structure to the applied excitation force, with respect to frequency 'ω'.

$$\mathbf{H(\omega)} = \frac{X(\omega)}{F(\omega)} \text{.....[equation 2.3]}$$

$$A(\omega, \phi) = A_0 \sin(\omega t + \phi) \text{.....[equation 2.4]}$$

The structure initially has to be maintained in free-free condition without any rigid support and therefore a continuous free flowing structure has infinite DOFs and

natural frequencies out of which a finite number of natural frequencies are obtained and studied. The FRF data are analyzed using different curve fitting procedures to derive the modal model of the structure. The various steps involved in modal testing are described in the later chapters. Experimental damping ratio (ζ) is calculated as,

$$\zeta = \frac{\Delta f}{2fn} \text{.....[equation 2.5]}$$

Bandwidth, $\Delta f = f_2 - f_1$ where f_1 and f_2 are called the half-power frequencies and they shift to $1/\sqrt{2}$ (or about 0.707 times) on either side of the peak amplitude, f1 is below f_n and f_2 is above fn.

2.2. Road load data acquisition (RLDA)

RLDA is the method used to record random-vibration data while the vehicle is being operated under different road conditions. The measured random-vibration data can later be used to build "vibration road-profiles" to test and validate future design iterations of similar vehicles. In RLDA, test vehicles are equipped with acceleration sensors placed strategically on the chassis. These sensors can record or sample multiple channels of acceleration data (Tx, Ty, Tz) and convert them into digital signals using an ADC (Analog to Digital Converter).

Random vibrations of vehicle ride are represented as Power spectral density (PSD) which is calculated to visualize energy distribution across the frequency spectrum and determine which frequencies are statistically more excited. PSD shows the average energy at a single frequency over time.

Power indicates mean-squared value of the signal (power), spectral across the frequency spectrum (as a function of frequency), Density Normalized to a frequency bandwidth. Units is g^2/Hz (for acceleration-time signal)

2.3. Whole Body vibrations (WBV)

The human and whole-body vibration solution is used to test and measure the effect of vibration on the human body. Whole-body and hand-arm vibration measurements are supported according to all relevant international standards - ISO 5349, ISO 8041, ISO 2631-1, and ISO 2631-5.

Table 2.1: Vibration Discomfort Levels and risk levels as per ISO 2631

a_w values	Comfort level
Less than 0.315 m/s2	Not uncomfortable
0.315-0.63 m/s2	A little uncomfortable
0.5-1 m/s2	Fairly uncomfortable
0.8-1.6 m/s2	Uncomfortable
1.25-2.5 m/s2	Very uncomfortable
Greater than 0.5 m/s2	Extremely uncomfortable
Human Health exposure limits	
VDV ($m/s^{1.75}$)	Description
<8.5	Health risks have not been objectively observed
8.5–17	Caution with respect to health risks is indicated
>17	Health risks are likely

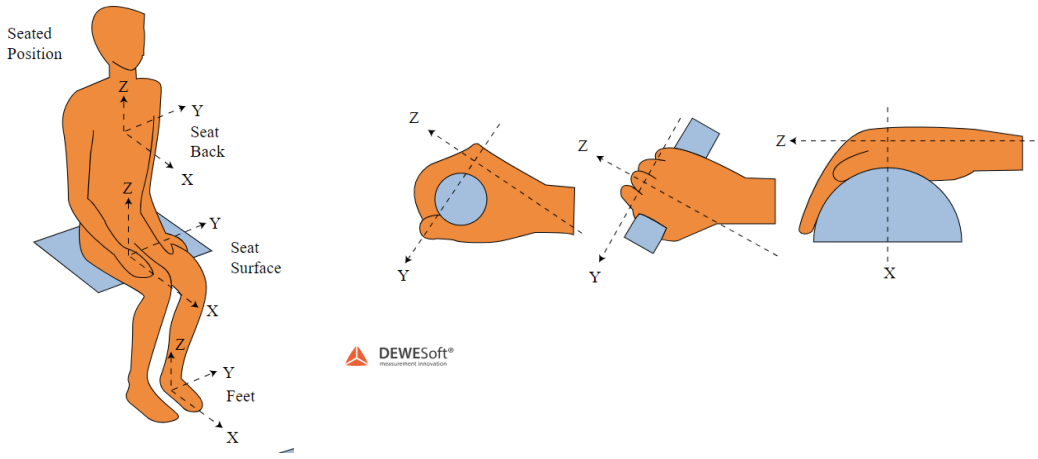


Figure 2.2 vibration measurement directions for WBV and HAV

2.4. Vehicle subsystems

2.4.1. Suspension System:

Spring: Provides elasticity to absorb shocks. The force exerted by a spring is given by Hooke's Law:

$F_s = k * x$…….[equation 2.1]

Fs= force exerted by the spring (N)

k= spring stiffness (N/m)

x= displacement from the equilibrium position (m)

CHAPTER 3

PROJECT METHODOLOGY

3.1 Problem definition

Experimental investigations have revealed, electric cargo 3-wheeler drivers are exposed to vibrations of amplitude more than 0.1 on a daily basis, leading to permanent back pain and hand muscles disorder which is above the safety limit for whole-body vibrations for humans as per ISO 2631. The critical mountings such as motor, steering column and driver seat are designed for ease and cost of manufacture. In the process, certain design decisions are taken just to pass certifications from government. But long run dynamic behaviours such as vibrations and its risks on driver health aren't studied. To address the problem, the research aims to maintain whole body and hand arm vibrations below 8 $m/s^{1.75}$ by introducing design modifications in front suspension and steering column that are easy to fabricate and costs less to manufacture.

3.2. Project Methodology

The Figure 3.1 describes the methodology followed for completing the project work, and various tools which were employed at various stages of project.

Task 1: Benchmarking of existing vehicles to select proto vehicle

Subtask-1: Identify 4 models of L5 category 3-wheelers and collect precise dimensions and specifications for documenting the dimensions of all models.

Task-2: Preparing simplified CAD model of selected proto vehicle.

Subtask-1: Repairing of overlapping surfaces and defeaturing holes, chamfers and blends in the final assembled model.

Subtask-2: Export the CAD model as .iges or .step format.

Task-3: FEA modal analysis

Subtask-1: Importing geometry and material assignment

Subtask-2: Applying Free-free boundary and mesh convergence.

Subtask-3: Solution and post processing of modal shapes and natural frequencies.

Task-4: Experimental Modal Analysis

Subtask-1: Maintaining the vehicle chassis in free-free condition and mounting of tri-axial accelerometers at critical locations.

Subtask-2: Setting-up the DAQ system and excitation of structure to obtain frequency response of the vehicle chassis.

Subtask 3-: Perform FFT analysis On DEWESoft to get amplitude v/s frequency plots and generate mode shapes and identify significant failure modes.

Task-5: Road Load Data Acquisition

Subtask-1: Prepare design of experiments for data collection in ride conditions such as road roughness, load, and vehicle speed varied in three levels. Mount ICP sensors on all locations of vibration transmissibility such as steering column, handle bar, seat mount, motor mounting.

Subtask-2: Set-up the DEWESoft DAQ for data collection and storing of RMS accelerations of vehicle chassis in x, y and z directions.

Subtask 3: Data processing of lower back and handle bar vibrations to calculate VDV values experienced by driver.

Task 6- Optimization of factors and responses using MiniTAb Software by importing CSV files of responses obtained.

Subtask-1: perform ANOVA for identifying significant influencing factor based on F-ratio values and analysing main-effects plot

Subtask-2: Perform Regression analysis with R^2 value more than 50% to achieve a model that if good fit for prediction of change in each response for unit increase in factors.

Subtask-3: Surface response methodology for optimizing factors by defining objective function to minimize VDV values and fixing the constraints and co-efficients.

Task-7: Conceptual design of cabin after optimization of obtained data

- Subtask-1: Using the processed data to create Prediction model of vibration transmissibility to create design constraints.

Task 8: **Design and simulation**

- Subtask-1: CAD modelling of the design in SIEMENS NX using the obtained optimized parameters.
- Subtask-1: Modal analysis simulation in ANSYS software

Task-9: Model assessment

- Subtask-1: Prepare prototype of modified parts and components
- Subtask-2: Replace old parts with new modified ones
- Subtask-3: Conduct modal analysis and road test with driver inside cabin and check if vibrations are under 8 $m/s^{1.75}$
- Compare old design and new design.

Task-6: Repeat optimization and design modification if vibration levels doesn't go below 8 $m/s^{1.75}$

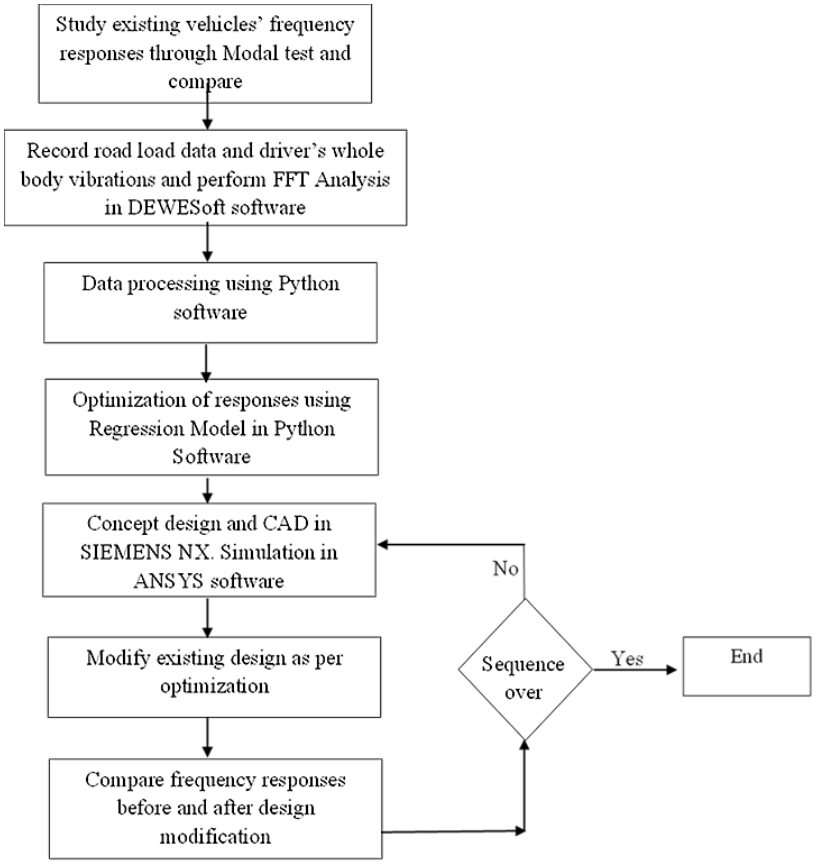


Figure 3.1 Project Methodology

CHAPTER 4

VEHICLE CHARACTERISTICS

4.1. Benchmarking of existing vehicles for Simulation

The benchmarking of existing vehicles is done before selecting the specimen vehicle for the experimental investigation. Table 4.1 shows the models of vehicles and their specifications. Figure 4.1 shows the vehicles that were used for benchmarking. Finally model A was chosen as the specimen vehicle due to reasons of easy availability and ownership of the company. The vehicle also had all typical design characteristics to understand the vibration characteristics of the L5 category electric three-wheeler cargo vehicle.

Table 4.1: Benchmarking of vehicles

Model Specification	Model : A	Model : B	Model : C	Model : D
Model make	Greaves ELTRA	Greaves e-Pro	Mahindra ZOR EV	Piaggio EV
Dimension (l×b×h in mm^3)	2978×1452×1866	2720×1305×1750	2940×1465×1840	3063 ×1422 ×1849
Kerb weight (kg)	480	420	420	367
Gross weight (kg)	800	710	730	690
Wheel base (mm)	2005	1818	1920	2025
Suspensions and steering geometry	Front: Anti-dive leading link Helical compression spring Hydraulic shock absorber Rear: Swinging arm Rubber compression spring Hydraulic shock absorber.			



Figure 4.1 (left to right) Model A to model D

4.2. Critical mountings and their natural frequencies

4.2.1. Motor and battery mountings

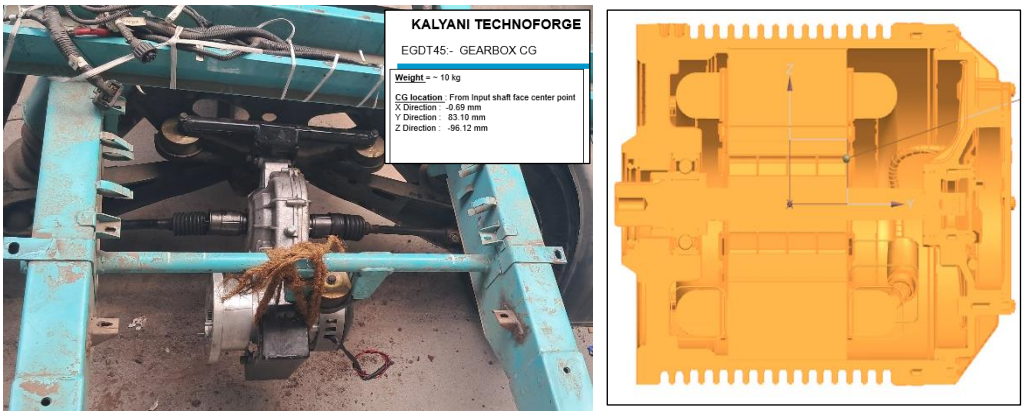


Figure 4.2 a) Kalyani Technoforge Transmission unit b) Motor section view

The motor mass properties extracted from SIEMENS NX mass properties is shown in figure .The motor make is KALYANI Technoforge E-motor weighing 16.87 kg with 6000 RPM. Therefore, the natural frequency of motor during operation is given by,

$$fn = \frac{\text{RPM}}{60} \text{.....[equation 3.1]}$$

$$fn = \frac{6000}{60} = 100 \text{ Hz.....[equation 3.2]}$$

It is evident from above relation that the vehicle components having natural frequencies in torsional and flexural directions same as motor mounting must be avoided in the bandwidth of 100 Hz to avoid resonance during ride condition.

CHAPTER 4

SIMULATION

4.1. Finite Element Analysis (FEA)

4.1.1. Prep-processing

The existing 3D CAD model of the vehicle is originally created with SIEMENS NX, as shown in Figure 4.1. The complex vehicle level CAD model undergoes geometric clean-up to remove unnecessary fillets, chamfers, holes, etc., to prepare it for analysis. The defeatured CAD model retaining the critical geometries and mass and stiffness properties and dimensions as per benchmarked values is shown in Figure 4.2.



Figure 4.1 Complex 3D CAD of vehicle in SIEMENS NX

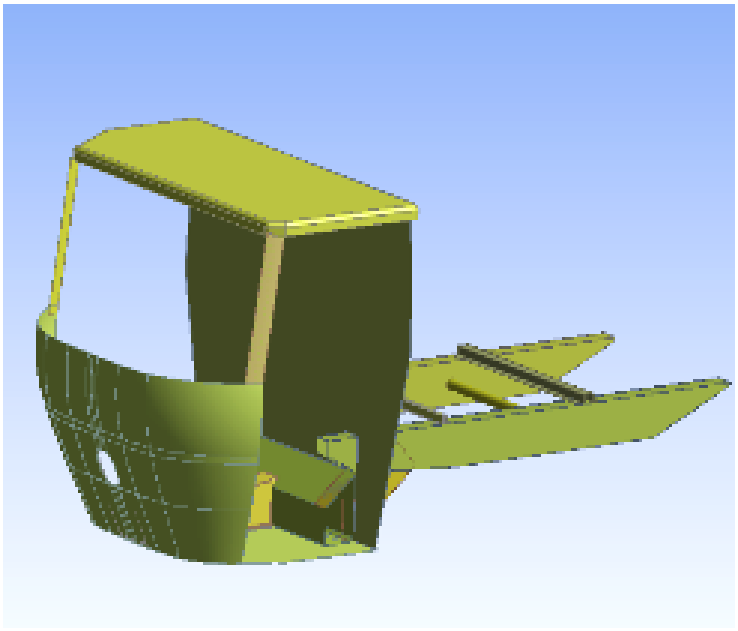


Figure 4.2 Repaired defeatured CAD imported for analysis

4.1.2. Mesh convergence study results

Methodology

The model is initially meshed using SOLID187 elements. SOLID 187 is a higher order 3-D, 10-node element characterized by quadratic displacement behavior, making it ideal for modeling irregular meshes. The element is defined by 10 nodes, each with three degrees of freedom: translations in the nodal x, y, and z directions. A mesh independence study is performed, using different element types namely Linear Tetrahedral, Quadratic Tetrahedral and Linear Hexahedral elements. The element sizes for each type of the element considered for mesh independence study are 17 mm, 12 mm, 10 mm and 8 mm. after which trial simulation is done. The mesh independence is obtained at 8 mm element size with first natural frequency 35 Hz.

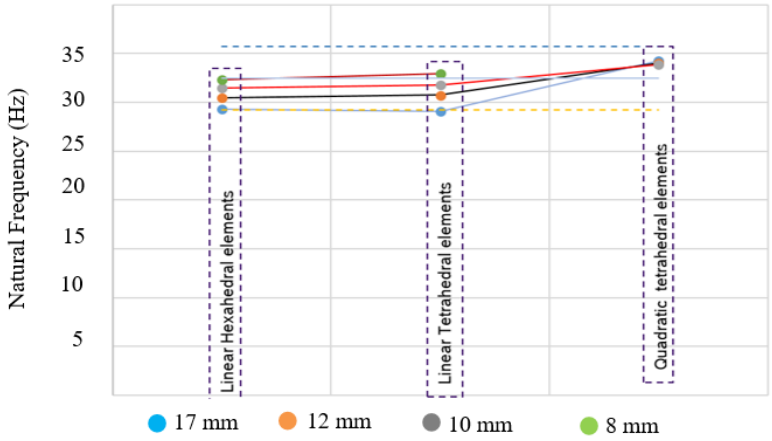


Figure 4.3 Mesh independence achieved at 8 mm element size

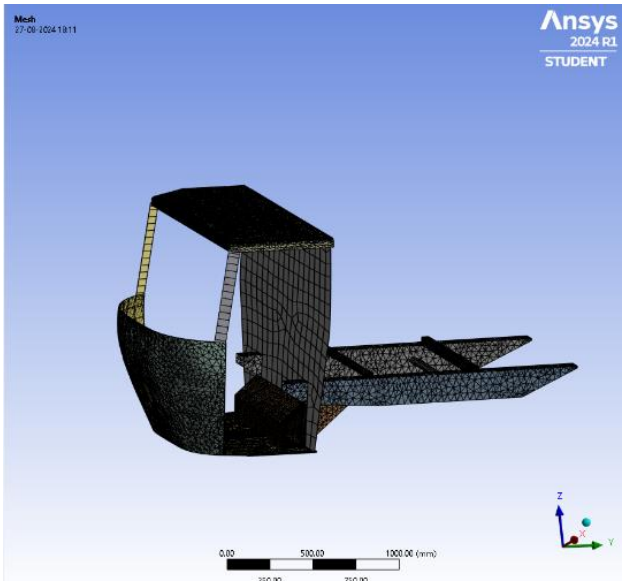


Figure 4.3 Optimized mesh for modal analysis with element size 8 mm

4.1.3. Material assignment

Table 4.1 shows material properties and center of mass of each component.

Materials properties of vehicle parts					
Part Name	Material	Density(kg/m3)	Tensile strength (MPa)	Young's Modulus (GPa)	Thickness(mm)
Driver cabin body	AISI 304 steel	7850	691	204	2
Chassis frames	SA 4130	7850	655	200	2.5
Motor mounting assembly	ASTM A48	7850	450	110	3
Battery cradle	IS:513, CR	7200	200	337	1.6
Chassis Cross members	IS: 3074, ERW2	8300	310	400	2.5
Steering column	SAE 4140	7900	355	452	2.6

4.1.3. Boundary conditions

The motor mounting(A), steering column (B),battery cradle (C) and seating (D)loaded on chassis with fixed support (G) and displacement constrained in Z direction but kept free in X and Y direction, at their respective center of mass positions as shown in Table 5.2.

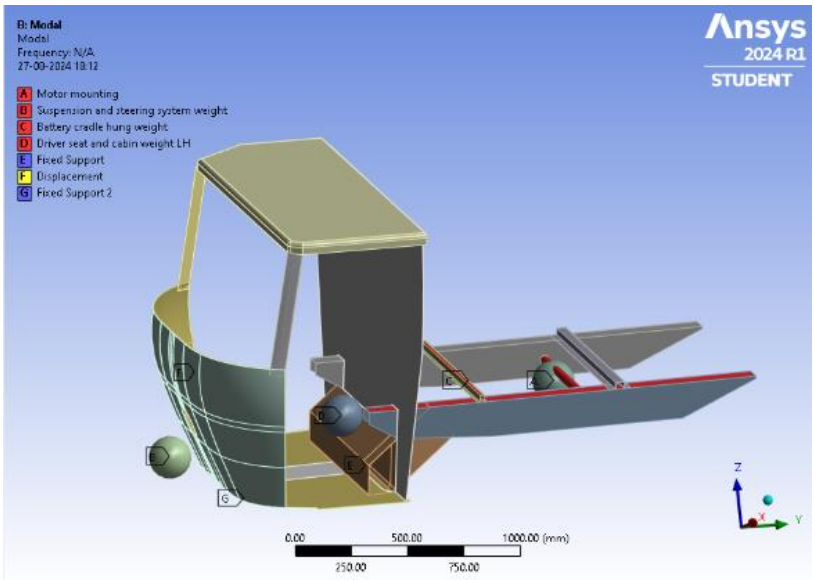


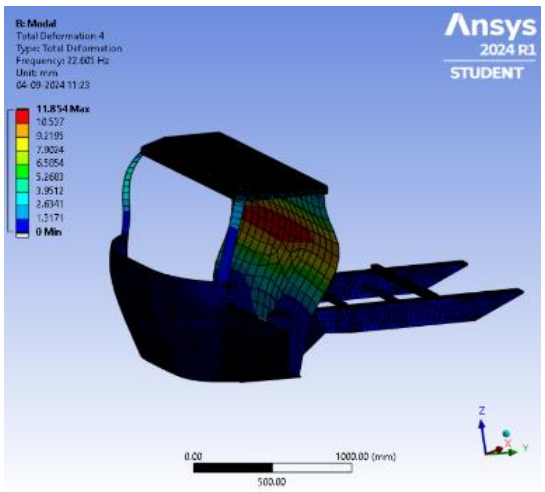
Figure 4.4 Boundary conditions and loads on vehicle chassis

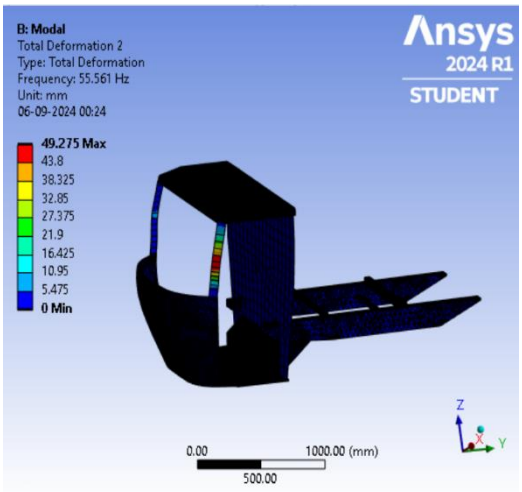
4.1.4. Solution and post-processing

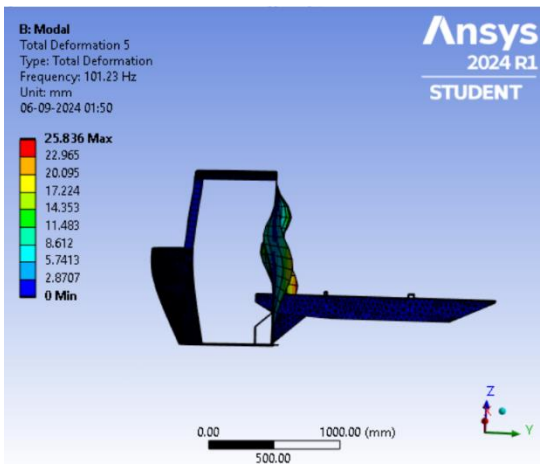
First five natural frequencies 1 to 6 are obtained as shown in Table 5.4 and respective mode shapes 1 to 6 are generated. Figure 4. Shows first two significant bending modes in vehicles Z-axis and Y-axis. with natural frequencies 22 Hz and 66 Hz. Figure 4. Shows first two significant torsional modes along Z-axis and Y-axis at 88 Hz and 101 Hz. Figure 4 c) shows combined bending and torsion mode at 101 Hz and 125 Hz. Figure 4.6 shows FEA modal results of steering column.

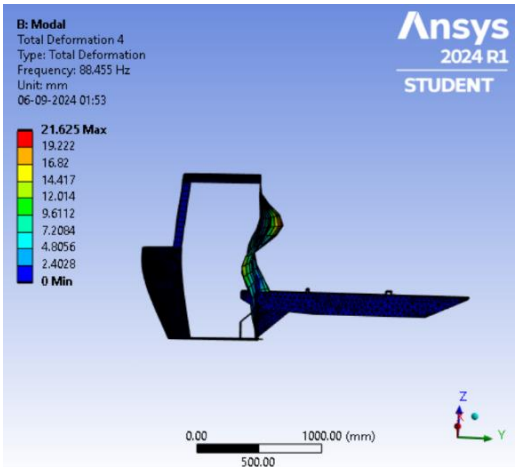
Table 5.4: Mode numbers Natural frequencies of vehicle structure

Mode number	Natural frequency (Hz)	Damping ratios
1	22.6	0.009
2	55.4	0.01
3	72.3	0.009
4	88.6	0.001
5	101	0.008
6	125	0.013









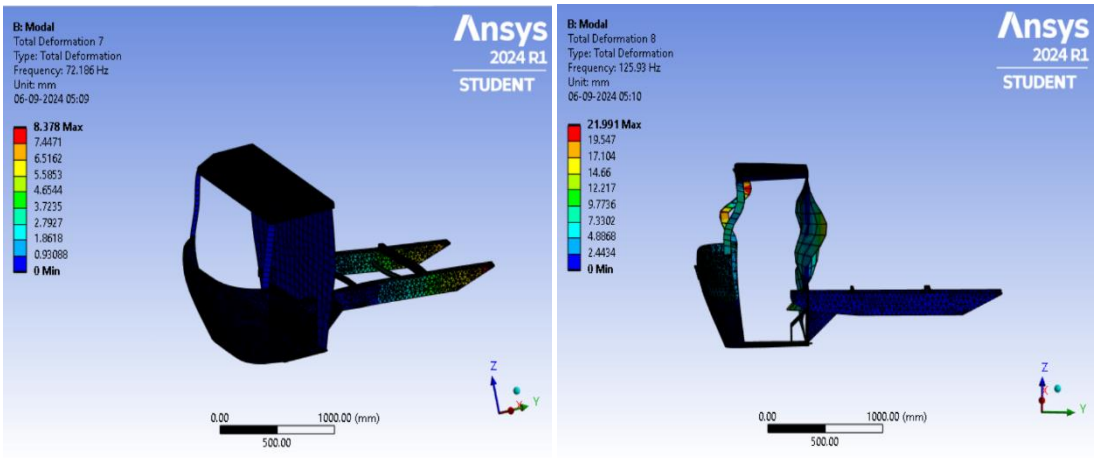


Figure 4.5 Mode shapes 1 to 6 and mode numbers graph

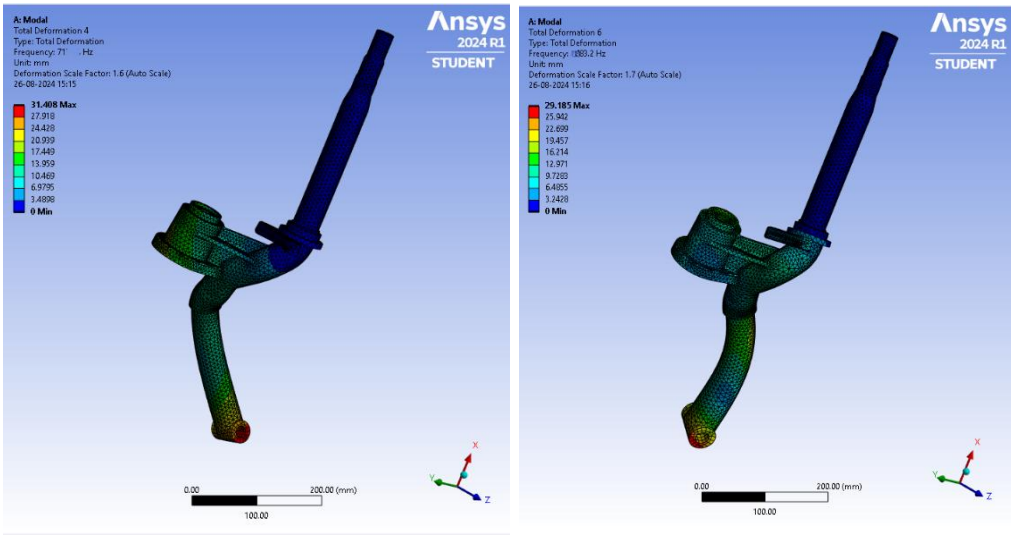


Figure 4.6 Two significant modes of steering column at 83 Hz and 119 Hz

CHAPTER 5

EXPERIMENTAL INVESTIGATION

5.1. Experimental Modal Analysis

5.1.1. Experimental setup

The cargo three-wheeler was tested in free-free condition. According to ISO 7626 guidelines the structure is supposed to be hanged with gently touching the ground. However, the three-wheeler was rested on its tires with only 50% tire pressure to mimic the hanging effect with co-ordinate system indicated as shown in the figure 5.2. The chassis and cabin with motor mounting and battery cradle were kept attached initially as shown in figure and later cargo container was attached later during testing with all other unnecessary components detached as shown in figure.

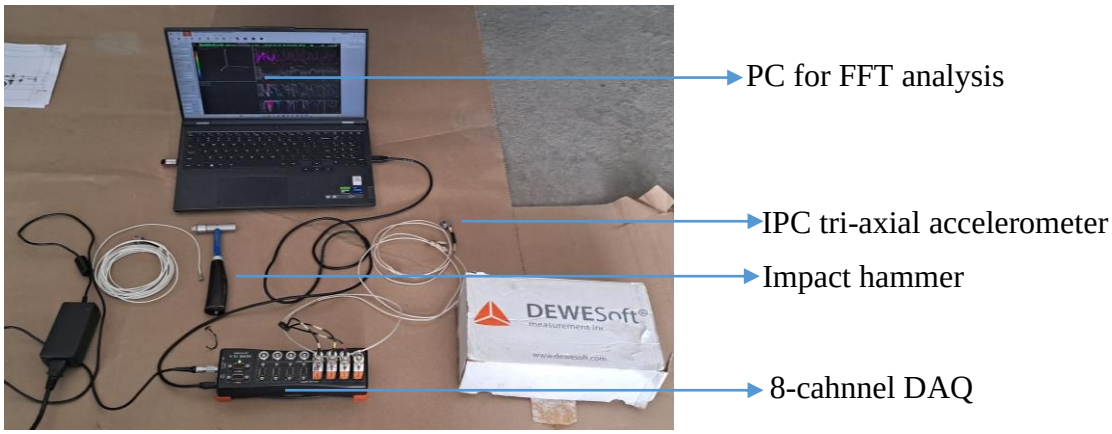


Figure 5.1: Experimental Modal Analysis set up with DEWESoft instruments



Figure 5.2. Vehicle chassis and cabin with 50% tire pressure

5.1.2. Structure excitation

An impact hammer with an integrated force transducer is used to excite the three-wheeler structure. A nylon tip is used, which excites the structure over a moderately wide frequency range. The impact hammer (PCB PIEZOTRONICS – D86D05) used has a sensitivity of 0.23mV/N (± 20%), a measurement range of ± 22240 N pk, and a mass of 0.32 kg.

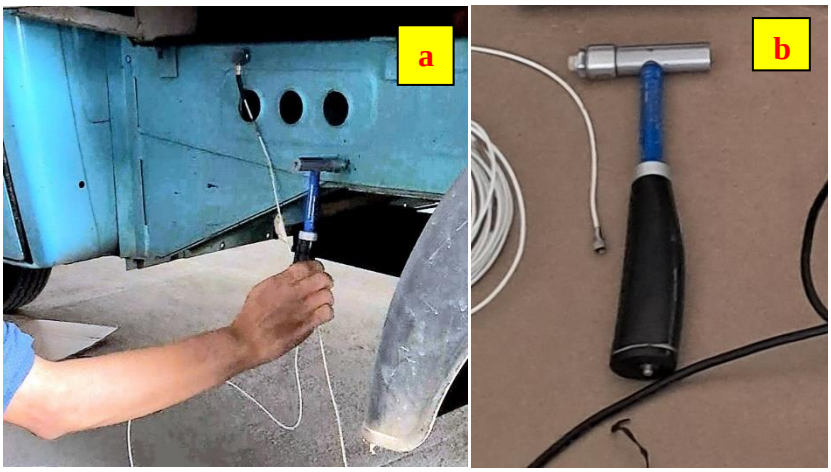


Figure 5.3 a) striking impact hammer on chassis member b) Impact hammer

5.1.3. Response measurement

The response of the structure to excitation is measured using an Integrated Electronics Piezo-electric (IEPE) accelerometer, which captures the acceleration response. Accelerometers with various sensitivity values are available, and the choice depends on the magnitude of vibrations in the structure. The tri-axial accelerometer (PCB PIEZOTRONICS – 356A15) used has a sensitivity of 100 mV/g, a measurement range of ± 50 g pk, and a frequency range of 2 Hz to 5 KHz (± 5%).

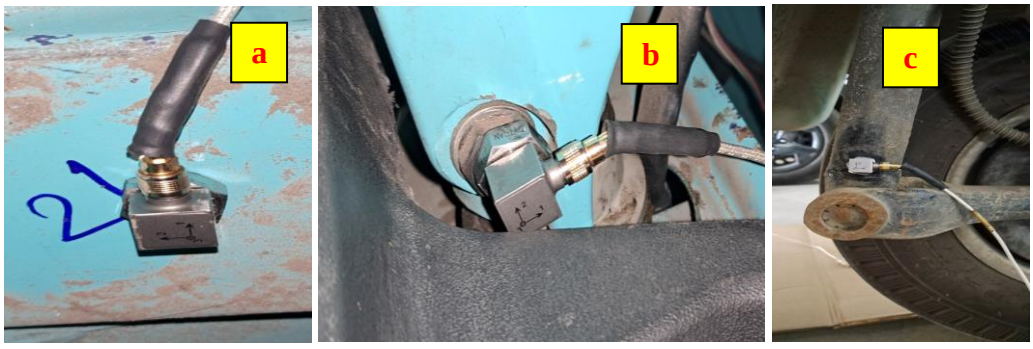


Figure 5.4. tri-axial accelerometers mounted a) beneath driver's seat b) on steering column top c) on steering column bottom near swing arm

5.1.4. Tap-test of vehicle chassis structure

The three-wheeler is free rested on ground. Through finite element analysis, the excitation and response locations are identified and marked on the three-wheeler. Figure 5.5 displays the 23 response measurement locations 1 to 9 on left chassis member, 10 and 11 on cabin floor and seat interface, 12 to 20 right chassis member, 21 beneath driver seat and points 22 and 23 on steering column top and bottom. The impact hammer and accelerometer are connected to the DAQ system, which is, in turn, connected to the computer. The DAQ system is configured in the software by inputting the sensitivities of the impact hammer and accelerometer, setting the pre-trigger, selecting the windowing type, and more. The accelerometer is mounted onto the three-wheeler using a magnetic base at the first marked response location, while the impact hammer excites the structure at the excitation location. The signals from the transducers are received and stored in the software.

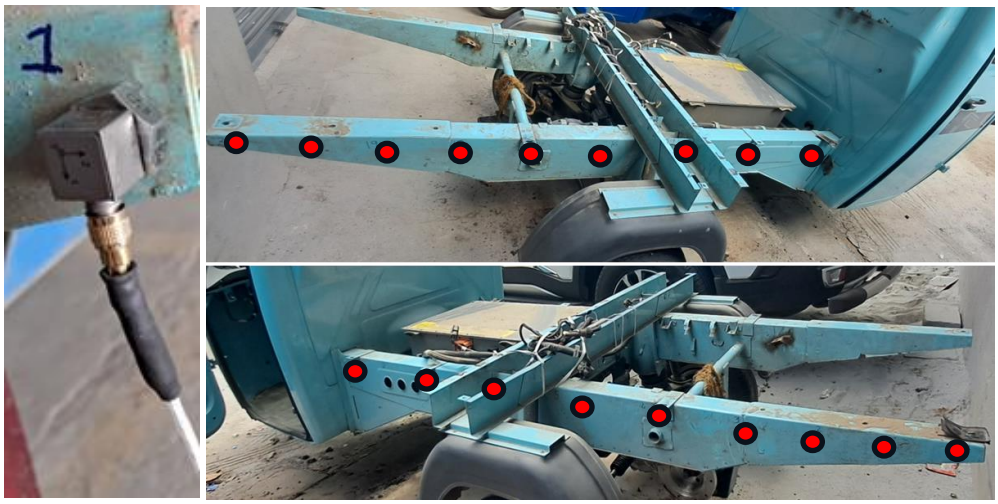


Figure 5.5 Locations of sensor mountings

The accelerometer is then moved to the next response location, and the structure is impacted at the same excitation location. This procedure is repeated until the response at all locations is captured. Good coherence between the impacts is ensured to achieve accurate mode shapes. This roving accelerometer technique results in one complete row of the FRF matrix, from which the proper mode shapes can be extracted.

5.1.5. Data acquisition and post-processing

The time domain signals from the force and response transducers are received and processed by a Data Acquisition (DAQ) system to convert them into frequency domain data using Fast Fourier Transform (FFT) analysis. The DAQ system is

connected to a computer equipped with the post-processing DEWESoft software package. The software allows for creating or importing the geometry of the structure as shown in figure and marking the response and excitation locations along with their axes orientation on the structure. The software settings and DEWESoft software user interface is shown in figure for processing EMA data. Modal parameters and shape functions in the form Operating Deflection Shapes (ODS) are estimated from the generated FRF using different curve fitting techniques and mode indicator functions provided in the software. All response measurement and excitation locations are marked on the model along with their axes orientation. Each of these points, along with their axes, forms a DOF of the model. Figure 7 shows the model of the three-wheeler, where points marked in red indicate the response measurement locations. From the measurement data, the software generates the frequency response function at each DOF, which is then processed to obtain the modal properties of the structure. A complex mode indicator function (CMIF)is used to identify the proper modes in the structure. The plot of the CMIF shows peaks at the damped natural frequencies. Table 5.1 shows the experimental and numerical values of natural frequencies of the vehicle structure. The percentage error is also calculated and lies with 5% difference which is permissible.

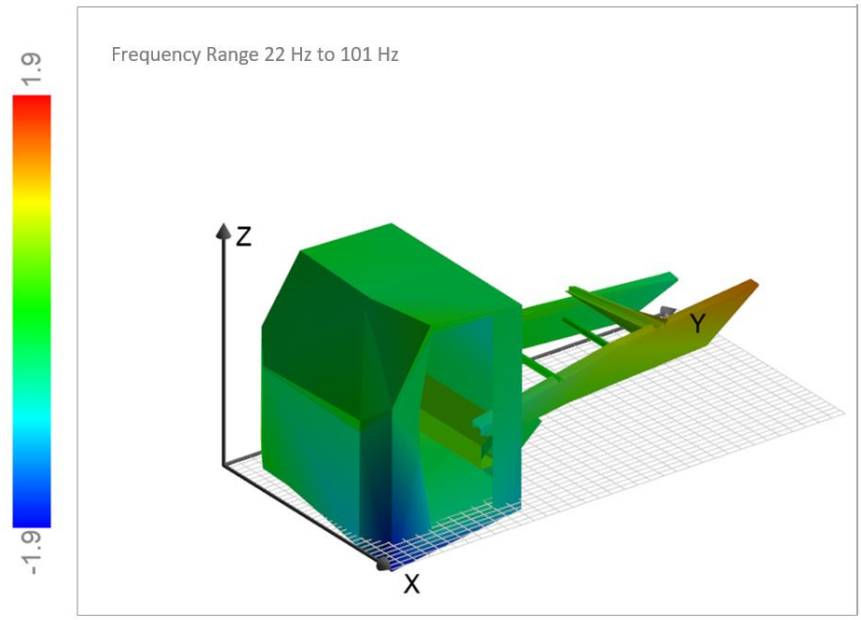


Figure 5.6 a) ODS of mode shapes generated in DEWESoft showing significant bending and torsion mode at 27 Hz and 101 Hz

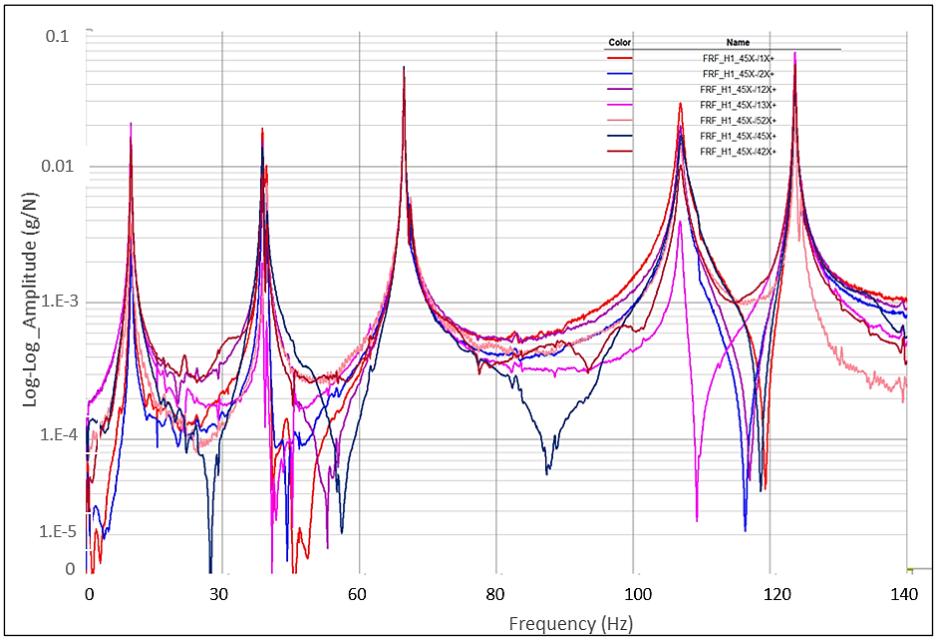


Figure 5.7 FRFs of vehicle natural frequencies

Table 5.1 Experimental and FEA results of modal analysis

Mode number	Natural frequency (Hz) FEA	Natural frequency (Hz) experimental	Difference in FEA and experimental values in %
1	22.6	20	8
2	55.4	50	10
3	72.3	68	5
4	88.6	105	18
5	101	122	13

5.2. Road Load Data Acquisition (RLDA)

5.2.1. Experimental setup and Vehicle Condition

Experiment setup inside cabin with PC encircled in orange and DAQ system encircled in red is shown in figure 4.5 (a). The tri-axial accelerometers are mounted on vehicle as shown in figure 4.5 (b). The 4 channel DAQ system with 3 responses input from accelerometer and vibration response is fed into software.

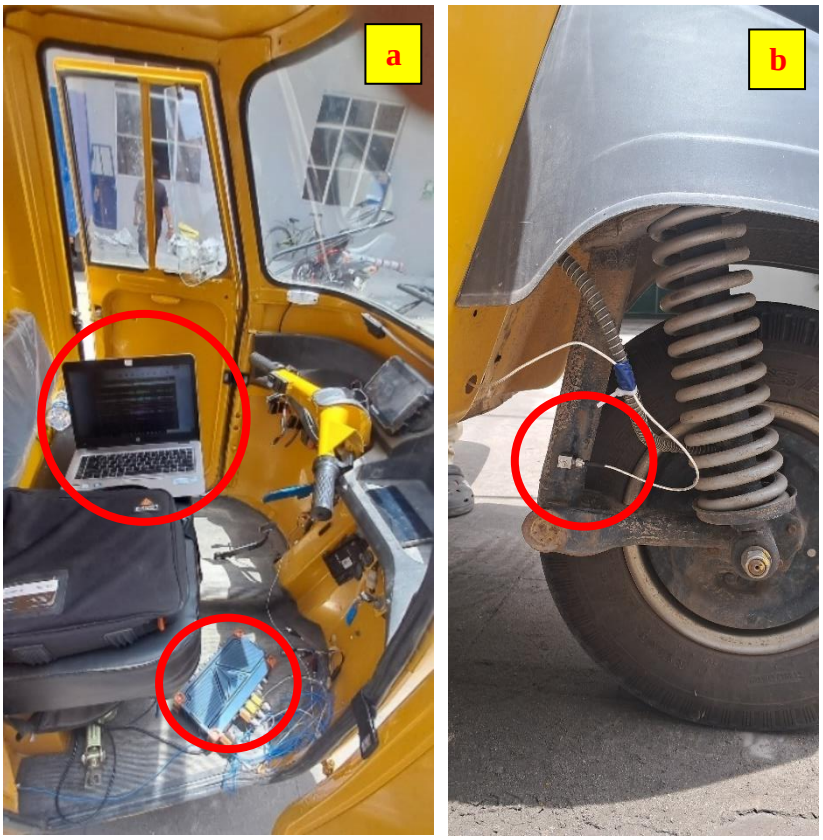


Figure 5.8 a) DEWESoft instruments setup for RLDA b) accelerometers mounted on steering column

5.6. Whole Body vibrations (WBV) test

5.6.1. Experimental setup

Experiment setup is shown in figure. one tri-axial accelerometer is integrated inside a 3D printed adapter as shown in figure to be placed on driver seat to for him to sit on top of it during ride. The 4 channel DAQ system with 3 responses input from accelerometer and vibration response is fed into software.

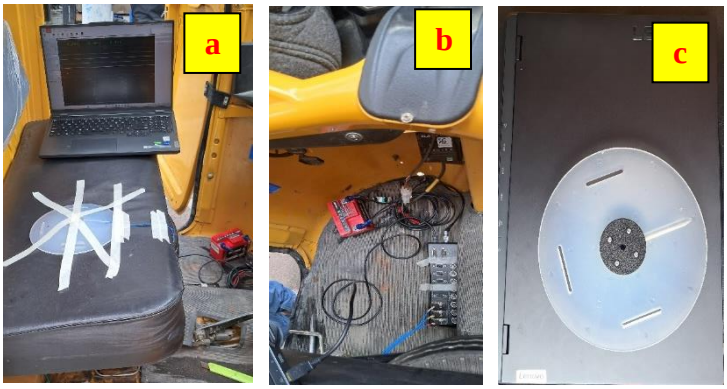


Figure 5.9 a) WBV setup in cabin b) Auxiliary battery and DAQ c) 3D printed adapter for seat-pad accelerometer

5.6.2. Ride conditions

Road roughness

Road profiles with roughness index (RI) 1.2 m/km considered as smooth, 3 m/km as medium and 5m/km as bad were chosen in surrounding areas of Kanakapura Main road for easy access from Greaves Electric Mobility company facility. The Ris are obtained from Bangalore Development Authority (BDA) 2024 database.



Figure 5.10 a) RI= 1.2 m/km) b) RI = 3 m/km c) RI = 5 m/km

Loading on vehicle

Trial run with no load condition was done for pre-test. The loads were gradually increased from level 1 150 kg, level 2 250 kg and level 3 390 kg. The standard sized sand cans each weighing 13 kgs were used in multiple numbers to achieve desired load value in the cargo container. Figure 5 (a (b) and (c) show no load condition,150 kgs load and 35 kmph vehicle speed respectively.



Figure 5.11 a) No load b) 150 kg load c) 35 kmph reading of vehicle

Vehicle speed

The vehicle three different average speed ranges from 35 kmph being in eco mode, 42 kmph being power-I mode and 50 kmph being in power-II mode. Figure 5. Shows vehicle in eco mode of 35 kmph.

CHAPTER 6

DESIGN OF EXPERIMENTS

6.1. Minitab V19 statistical used for DOE

Table .shows factors and corresponding variability levels. Table shows DOE based on Taguchi design. A 3-factor 3-level L27 array is used for a comprehensive collection of data which makes optimization more accurate. The L27 array is chosen because of constraints of time, availability of vehicle, driver, DAQ instruments, cost of experiments and traffic conditions.

Table 6.1. Factor and levels table

Factors	Level 1	Level 2	Level 3
Road roughness (m/km)	1.2	3	5
Vehicle speed (kmph)	35	42	50
Loading (kg)	150	250	390

Table 6.2. L29 orthogonal array using Taguchi method

Run	Road Roughness (m/km)	Average Vehicle Speed (kmph)	Loading (kg)
1	1.2	35	150
2	1.2	35	250
3	1.2	35	390
4	1.2	42	150
5	1.2	42	250
6	1.2	42	390
7	1.2	50	150
8	1.2	50	250
9	1.2	50	390
10	3	35	150
11	3	35	250
12	3	35	390
13	3	42	150

14	3	42	250
15	3	42	390
16	3	50	150
17	3	50	250
18	3	50	390
19	5	35	150
20	5	35	250
21	5	35	390
22	5	42	150
23	5	42	250
24	5	42	390
25	5	50	150
26	5	50	250
27	5	50	390

6.1.1. RLDA results

The power spectral densities ranged from 0.02 to 0.072 g^2/Hz. The RMS accelerations were derived for each of 27 experiments. Figure 6.1 shows the consolidated RLDA log-log plot of PSD v/s Frequency for road with RI 1.2 m/km. The PSD values range from 0.02 to 0.03 g^2/Hz and lie in frequency range 0 to 30 Hz. For road with RI 3 m/km, PSD values range from 0.03 to 0.056 g^2/Hz in frequency range 34 Hz to 67 Hz and road with RI 5 m/km PSD range from 0.055 to 0.075 and lie in 70 Hz to 147 Hz. Table 6.3 shows isolated PSD responses of seat-pad and handle bar vibrations.

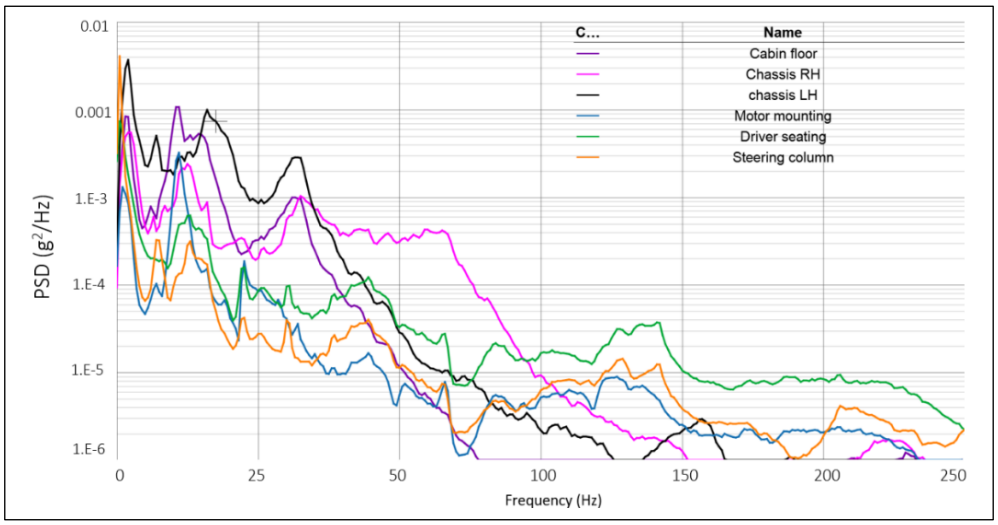


Figure 6. Consolidated PSD v/s frequency log-log plot for all 3 road roughness

conditions

Table 6.3 RMS accelerations and PSD values

Frequency [Hz]	Z_w/CPB [m/s2]	Y_w/CPB [m/s2]	X_w/CPB [m/s2]	PSD [$m/s^{1.75}$]
1	0.0077	0.0082	0.0018	1
1.25	0.0106	0.0115	0.0035	0.5
1.6	0.0102	0.0277	0.0019	0.5
2	0.0132	0.0046	0.0079	0.5
2.5	0.0016	0.0079	0.0031	0.5
3.15	0.0108	0.0028	0.0003	0.5
8.8	0.0126	0.0023	0.0004	0.5
16.3	0.1435	0.0337	0.0279	0.5
21	0.1503	0.0142	0.0086	0.5
22	0.0188	0.0137	0.0053	0.5
22	0.2783	0.0332	0.028	1
22.7	0.0601	0.0116	0.0018	2.2
23	0.0436	0.0106	0.0024	4.6
24	0.0945	0.0374	0.007	3.8
25	0.0743	0.0152	0.0037	3.12
31.5	0.0221	0.0066	0.0072	2
40	0.1522	0.0547	0.1085	4
50	0.101	0.0147	0.0443	2
63	0.0525	0.0078	0.0148	4.3
80	0.0421	0.0029	0.0082	3.2
27	0.0945	0.0374	0.007	3.8
25	0.0743	0.0152	0.0037	3.12
31.5	0.0221	0.0066	0.0072	2
40	0.1522	0.0547	0.1085	4
54	0.101	0.0147	0.0443	2
63	0.0525	0.0078	0.0148	4.3
80	0.0421	0.0029	0.0082	3.2

6.1.2. Vibration transmissibility

The vibration transmissibility shows the percentage of vibrations from a source transmitted to second body in contact. The isolation of vibrations can be done effectively by calculating the transmissibility. At all ride conditions the transmissibility value remains same since the components are fixed in a vehicle unlike multi state mechanisms that exist in complex machines. The output vibration of first body is given as input to the second body. Then, transmissibility is calculated as the ratio of vibration at output to vibration at input. Equation shows transmissibility of a system.

$$\boldsymbol{T = \frac{a_{output}}{a_{input}} = \frac{1}{\sqrt{(1-r^2)^2+(2\xi r)^2}}} \quad \textbf{……..[equation 6.1]}$$

where:

a_{output} is RMS acceleration captured by accelerometer from 1st component

a_{input} is RMS acceleration captured by accelerometer at interface of 1st and 2nd componenet.

$r = \frac{f}{f_n}$ *is the frequency ratio*

(ξ) *is the damping ratio of the system.*

Using above relation, vibration transmissibility is calculated at chassis -seat interface using seat pad accelerometer and at steering column bottom and top of hand-arm interface. The calculations are as below.

T at motor mount-seat interface, $\boldsymbol{T}\text{ms} = \frac{A_{output}}{A_{input}} = \frac{0.023}{0.025} = 0.92 = 92\%$

T at steering column-handle bar interface, $\boldsymbol{T = \frac{A_{output}}{A_{input}} = \frac{0.021}{0.029} \approx 0.724 = 72\%}$

5.1.2. WBV testing results

The vibration dose values ranged from 3 to 17 for 27 responses. Figure 6.2 shows DEWESoft software results of whole body vibrations and hand-arm vibrations.

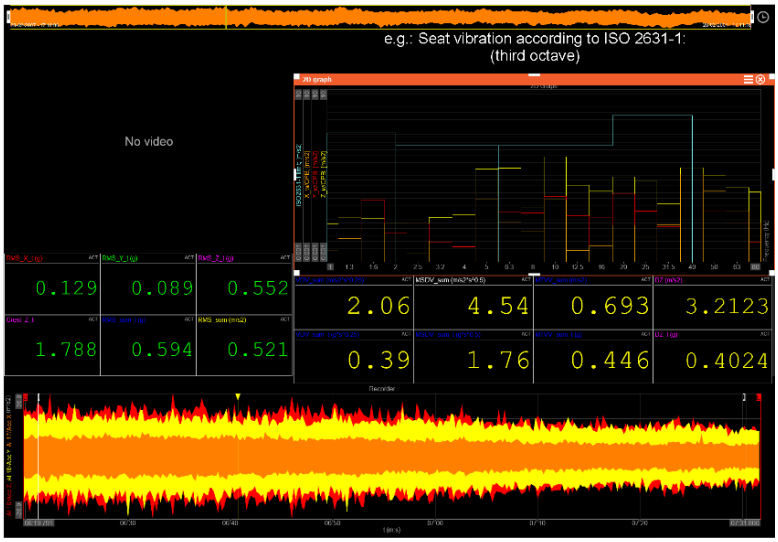


Figure 6.2 DEWESoft WBV processing module results

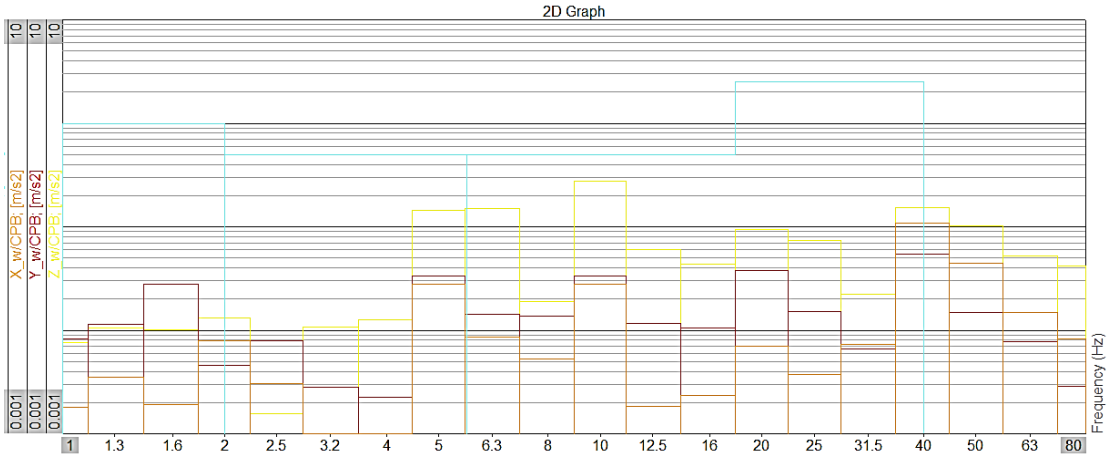


Figure 6.3 amplitude v/s frequency plots for lower-back and hands.

Vibration Dose values for above responses is derived using equation (6.1). figure shows amplitude v/s frequency plot.

$$\mathbf{VDV} = \left(\int_T [W(t) \cdot a(t)]^4 \, dt\right)^{1/4} \text{.....[equation 6.1]}$$

a(t) is the acceleration in the time domain.

W(t) is the frequency weighting function.

T is the duration of the measurement**.**

Further, VDV v/s road roughness index is plotted. Figure 6.4 shows VDV in $m/s^{1.75}$ varying rapidly with increase in road roughness index from 1.2 (smooth road). The plot clearly indicates the severity of VDV crossing 8.5 $m/s^{1.75}$in road roughness of 2.5 and above.

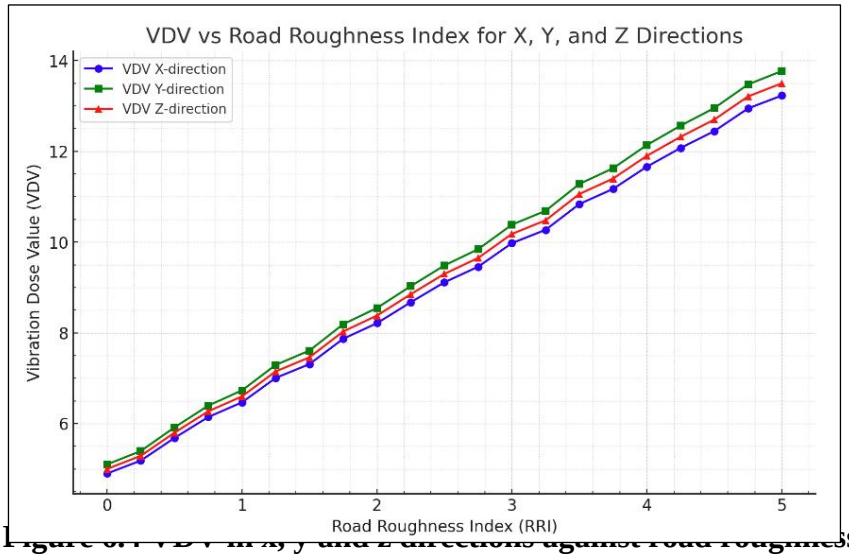


Figure 6.4 VDV in X, Y and Z directions against road roughness

5.2. Responses obtained

Table shows 27 responses of PSD, RMS acceleration of lower-back and hand-arm and VDV values of lower-back and hand-arm. The PSD ranges from 0.02 to 0.072, VDV of lower back ranges from 5 $m/s^{1.75}$ to 13. 5 $m/s^{1.75}$ and VDV hand-arm ranges from 3 to 8.3 $m/s^{1.75}$. These values are beyond safe limits as per ISO 2631.

Table 6.4 responses obtained after experiment

Run	PSD (g^2/Hz)	RMS Acceleration (Lower Back) (m/s^2)	RMS Acceleration (Hand-Arm) (m/s^2)	VDV (Lower Back) ($m/s^{1.75}$)	VDV (Hand-Arm) ($m/s^{1.75}$)
1	0.02	0.1	0.08	5	3
2	0.022	0.12	0.09	5.2	3.1
3	0.024	0.14	0.1	5.5	3.3
4	0.026	0.15	0.12	6	3.5
5	0.028	0.17	0.13	6.3	3.7
6	0.03	0.18	0.14	6.5	3.8
7	0.032	0.2	0.15	7	4
8	0.034	0.22	0.16	7.3	4.2
9	0.036	0.24	0.18	7.5	4.3
10	0.038	0.25	0.2	8	5
11	0.04	0.27	0.22	8.3	5.2

12	0.042	0.28	0.23	8.5	5.3
13	0.044	0.3	0.25	9	5.5
14	0.046	0.32	0.27	9.3	5.7
15	0.048	0.34	0.28	9.5	5.8
16	0.05	0.35	0.3	10	6
17	0.052	0.37	0.32	10.3	6.2
18	0.054	0.38	0.33	10.5	6.3
19	0.056	0.4	0.35	11	7
20	0.058	0.42	0.37	11.3	7.2
21	0.06	0.44	0.38	11.5	7.3
22	0.062	0.45	0.4	12	7.5
23	0.064	0.47	0.42	12.3	7.7
24	0.066	0.48	0.43	12.5	7.8
25	0.068	0.5	0.45	13	8
26	0.07	0.52	0.47	13.3	8.2
27	0.072	0.54	0.48	13.5	8.3

5.3. Analysis of Taguchi design responses

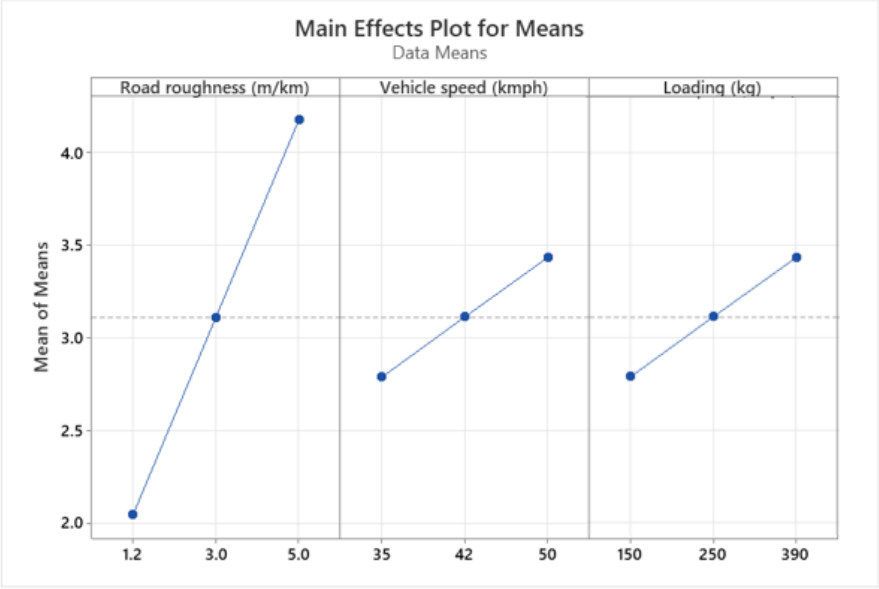


Figure 6.5 shows main effects plot for means of means v/s three factors

Table 6.5 Response Table for Signal to Noise Ratios: Smaller is better

Level	Road roughness (m/km)	Vehicle speed (kmph)	Loading (kg)
1	-10.15	-12.35	-13.18
2	-13.71	-13.40	-13.30
3	-16.21	-14.31	-13.58
Delta	6.06	1.96	0.40
Rank	1	2	3

5.3. Analysis of variance (ANOVA)

The calculations of ANOVA can be characterized as computing a number of means and variances, dividing two variances and comparing the ratio to an F-distribution The calculations done in excel is shown in appendix A. Table value to determine statistical significance.

The value for mean square is determined by:

$$\boldsymbol{MeanSquare(MS)value}\mathbf{MS} = \frac{\textbf{Sum of squares (SS)}}{\textbf{Degrees of freedom (Df)}} \textbf{.....[equation 5.2]}$$

The F-ratio is determined by:

$$\mathbf{F-ratio, F} = \frac{\mathbf{MS_{between\ groups}}}{\mathbf{MS_{within\ groups}}} \textbf{.....[equation 5.3]}$$

5.3.1. ANOVA Results

1. **PSD ($m^2/s^3/Hz$)**

   Road Roughness: F = 19146.79, p < 0.001

   Vehicle Speed: F = 2126.23, p < 0.001

   Loading: F = 234.43, p < 0.001

2. **RMS Acceleration (Lower Back) (m/s^2)**

Road Roughness: F = 13838.40, p < 0.001

Vehicle Speed: F = 1537.86, p < 0.001

Loading: F = 190.06, p < 0.001

3. **RMS Acceleration (Hand-Arm) (m/s^2)**

Road Roughness: F = 7472.36, p < 0.001

Vehicle Speed: F = 741.04, p < 0.001

Loading: F = 66.91, p < 0.001

4. **VDV (Lower Back) ($m/s^{(1.75)}$)**

Road Roughness: F = 15679.65, p < 0.001

Vehicle Speed: F = 1753.79, p < 0.001

Loading: F = 105.74, p < 0.001

5. **VDV (Hand-Arm) ($m/s^{(1.75)}$)**

Road Roughness: F = 14476.27, p < 0.001

Vehicle Speed: F = 918.99, p < 0.001

Loading: F = 78.02, p < 0.001

All three factors (Road Roughness, Vehicle Speed, Loading) significantly affect each of the responses (PSD, RMS Acceleration, VDV) with p-values less than 0.001. Road Roughness has the largest impact on all responses, as indicated by the highest F-values.

5.4. Regression Analysis

Regression Analysis showed Road Roughness is the most significant factor affecting all responses, followed by Vehicle Speed and Loading. The high R-squared values (0.899) indicate that the models explain nearly all the variability in the data. Following regression equation for all responses prediction is given.

1. PSD ($m^2/s^3/Hz$):

$$\text{PSD} = -0.0212 + 0.0095 \times \text{Road Roughness} + 0.0008 \times \text{Vehicle Speed} + 0.00001651 \times \text{Loading} \quad \text{[equation 5.5]}$$

2. RMS Acceleration (Lower Back) (m/s^2):

$$\text{RMS Acceleration (Lower Back)} = -0.0154 + 0.0098 \times \text{Road Roughness} + 0.0007 \times \text{Vehicle Speed} + 0.000015 \times \text{Loading} \quad \text{[equation 5.6]}$$

3. RMS Acceleration (Hand-Arm) (m/s^2):

$$\text{RMS Acceleration (Hand-Arm)} = -0.0193 + 0.0075 \times \text{Road Roughness} + 0.0007 \times \text{Vehicle Speed} + 0.000013 \times \text{Loading} \quad \text{[equation 5.7]}$$

4. VDV (Lower Back) ($m/s^{1.75}$):

$$\text{VDV (Lower Back)} = -0.8462 + 1.1282 \times \text{Road Roughness} + 0.0737 \times \text{Vehicle Speed} + 0.0013 \times \text{Loading} \quad \text{[equation 5.8]}$$

5. VDV (Hand-Arm) ($m/s^{1.75}$):

$$\text{VDV (Hand-Arm)} = -0.7410 + 1.0545 \times \text{Road Roughness} + 0.0673 \times \text{Vehicle Speed} + 0.0012 \times \text{Loading} \quad \text{[equation 5.9]}$$

CHAPTER 7

OPTIMIZATION

7.1. Response Surface Methodology (RSM)

RSM is best suited to identify the optimal levels of each factor that minimize or maximize the desired response. In this case VDV values should be minimized below 1.5m/s^2 based on responses obtained. The objective function can be represented as a second-order polynomial equation based on the factors involved. Assuming X_1 is the road roughness, X_2 is the vehicle speed, and X_3 is the loading, the objective function Y (VDV) can be expressed as: is defined by equation (6.1)

Minimize $$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \beta_3 X_3 + \beta_{11} X_1^2 + \beta_{22} X_2^2 + \beta_{33} X_3^2 + \beta_{12} X_1 X_2 + \beta_{13} X_1 X_3 + \beta_{23} X_2 X_3 + \epsilon$$ **.....[equation 6.1]**

Where:

Y= Vibration Dose Value (VDV) or another measure of vibration.

β_0 = Constant term (intercept).

β_1,β_2,β_3 = Linear coefficients for road roughness, vehicle speed, and loading, respectively.

$\beta_{11},\beta_{22},\beta_{33}$ = Quadratic coefficients for road roughness, vehicle speed, and loading, respectively.

$\beta_{12},\beta_{13},\beta_{23}$ = Interaction coefficients between the factors.

ϵ = Error term accounting for variability not explained by the model.

Optimization of factors using above objective function can be achieved using following coefficients and constraints,

$\beta 0 = -0.8462$

$\beta 1 = 1.1282$

$\beta 2 = 0.0737$

$\beta 3 = 0.0013$

Constraints:

$1.2 \leq X_1 \leq 5$ (Road Roughness)

$10 \leq X_2 \leq 55$ (Vehicle Speed)

$100 \leq X_3 \leq 250$ (Loading)

The MS excel response surface module solver data is attached in appendix A for calculating the optimized factor values. The optimized results are obtained as follows

1. Road roughness: ranging from 2 to 4 m/km
2. Loading condition: up to 315 kgs
3. Average vehicle speed: 43 kmph

Using above conditions of ride the VDV values can be maintained below 4 m/s1.75 which implies RMS acceleration can be below 1.3 m/s^2. The Perato chart for factors v/s optimized values is shown in figure 7.1.

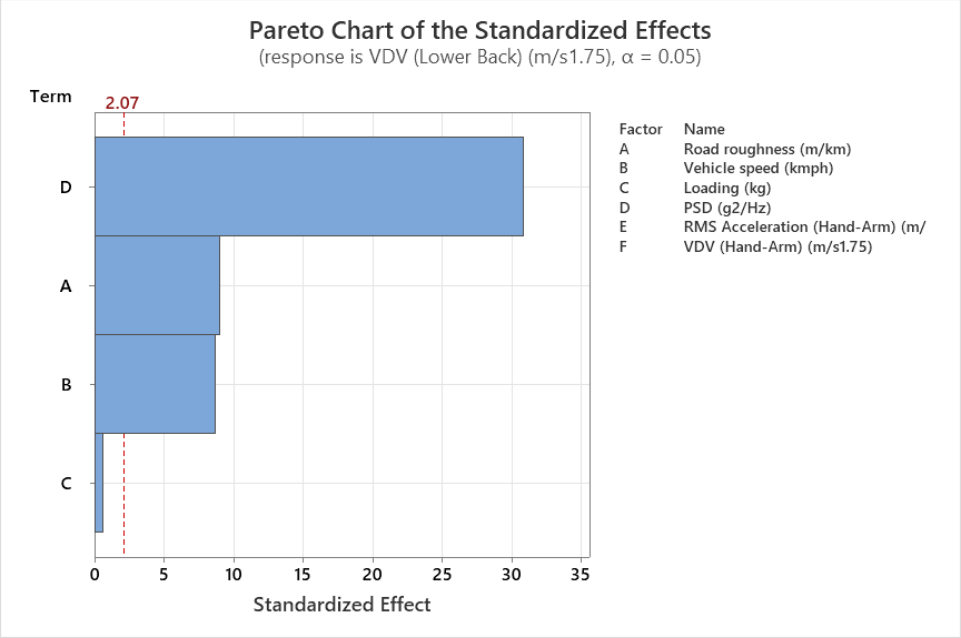


Figure 7.1 Pareto chart for factors v/s optimized values

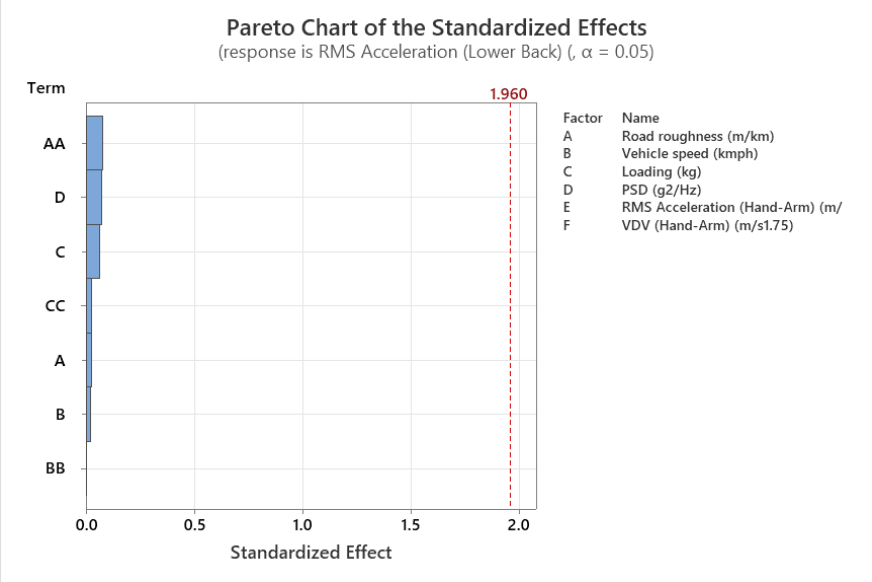


Figure 7.2 Pareto chart for combination of factors and effects

CHAPTER 8

DESIGN AND FABRICATION

8.1. Design inputs

In previous chapters the results of modal analysis, RLDA human vibration test and optimization make way for understanding influence of physical design features of cabin components on responses for which modification is proposed. The explanation for each factor variation and contributing components are discussed below:

1. Natural frequency of vehicle chassis and steering column lie in overlapping ranges leading to excitations during ride conditions. This demands modification in steering column geometry.

2. Responses isolated from driver seat show vibration transmissibility more than 69% from motor mounting through chassis member. This demands design modification of seat mounting and its position.

3. Identification of key Components Affected from Vibration Sources: Steering column and driver seat being two main transmissibility points, these components demand for design modification.

4. Connection Between Factors and Design:

Road Roughness: Need to modify the suspension system or add adaptive damping features.

Vehicle Speed: Changes in vehicle speed can affect the performance of shock absorber directly in cabin.

Loading: Adjustments might be required in structural elements or materials used to handle different loading conditions more effectively.

8.1.1. Scoring Matrix of design inputs

Research methodology contains various scoring matrix to rate the significance of any factor that needs to be given a priority to proceed further in product development with highly controlled resources and constraints.

Table 7.1 Product screening Matrix for type of steering column selection

Criteria	Front suspension				
	Trailing link				
	Trailing link	Leading link	Old Design	Telescopic fork	**U-fork**
Simplicity (Low grade = low implicity)	0,33	0,33	0,17	0,33	**0,25**
Space demand (Low grade = takes a lot of space)	0,33	0,33	0,17	0,25	**0,25**
Vibration transmissibility Handling (Low grade = poor handling)	0,67	0,89	0,22	0,67	**1,11**
Wobbling handling (Low grade = poor dynamics)	0,67	0,67	0,44	0,89	**0,67**
Economic aspect (manufacturing cost) (Low grade = high manufacturing costs)	0,78	0,78	0,78	0,97	**0,78**
Material selection (maintenance, durability) (Low grade = high costs for the customer)	0,78	0,78	0,58	0,78	**0,97**
Grade:	3,56	3,78	2,36	3,89	**4,03**
Result:	Discarded	Discarded	Discarded	Discarded	**Chosen**

8.2.U-fork steering column

8.2.1. 3D model

The modified geometry was designed in SIEMENS NX software. Figure 7.1 shows CAD model of U-fork steering column. Figure shows 2D draft with all dimensions in mm. Table shows list of parts.

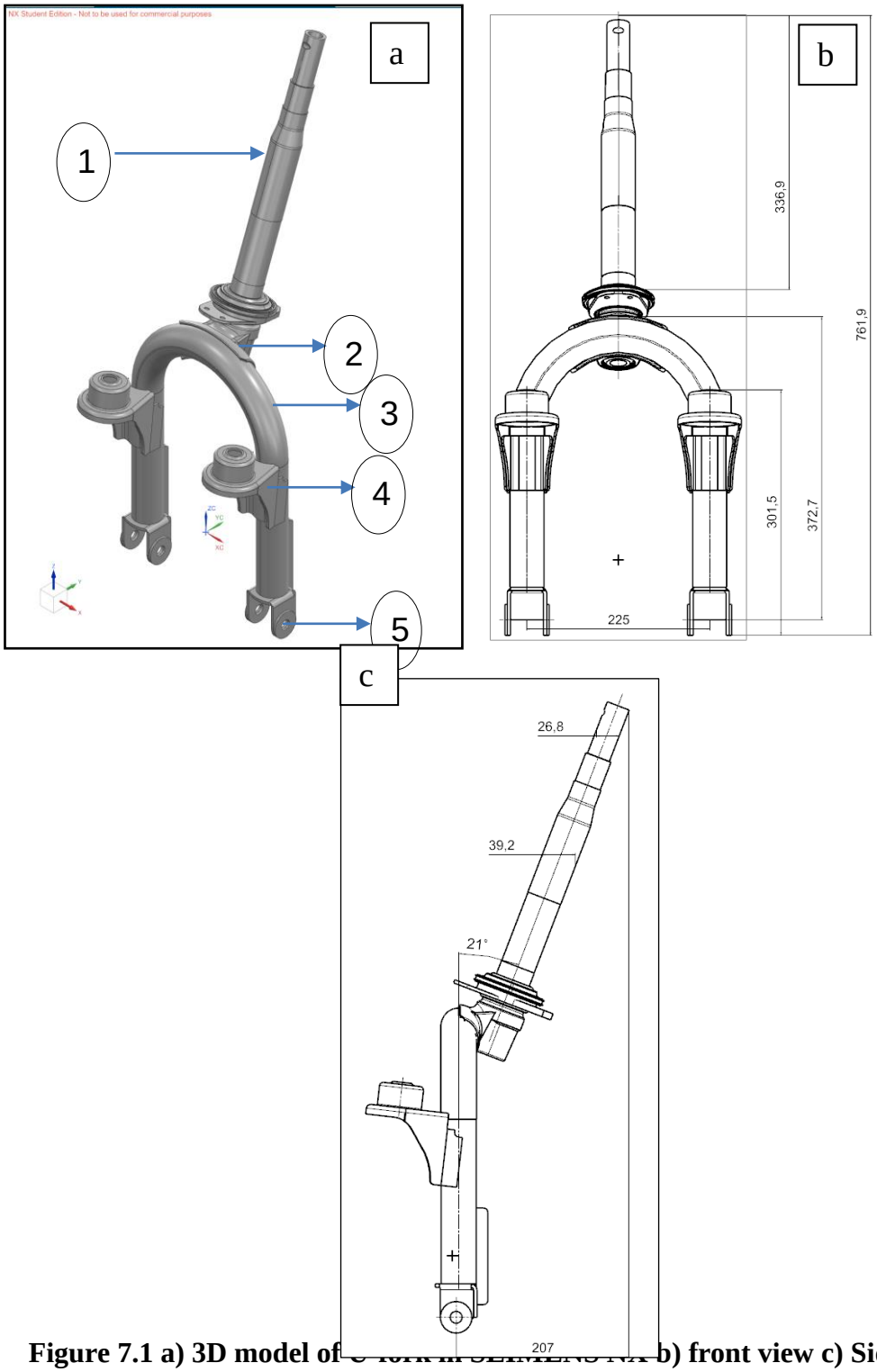


Figure 7.1 a) 3D model of U-fork in SIEMENS NX b) front view c) Side view

Table 7.2 list of parts and material data

Materials properties of Steering column					
Part Name	Material	Density(kg/m3)	Tensile strength(MPa)	Young's Modulus(GPa)	Thickness(mm)
U-fork	SAE 4310	7950	691	204	2
Weld bracket	AISI 4454	7850	655	200	2.5
Steering column	BSK 46	7850	450 Mpa	500 Mpa	3
Mounting bracket	IS:513, CR	7200	200 Mpa	337 Mpa	1.6
Spring seater	IS: 3074, ERW2	7800	310Mpa	400 Mpa	2.5

7.2.2 FEA modal analysis of U-fork steering column

Meshing and boundary conditions

The optimized mesh of SOLID 186 element size 8mm with jacobian index of 0.62 is shown in Figure 7.2 . Front axle weight of 182 kgs acts on mounting brackets, with steering column fixed in z and free in x and y directions. Mesh convergence was achieved at 12 mm element size. Figure shows mesh of 10 mm and 8 mm element size.

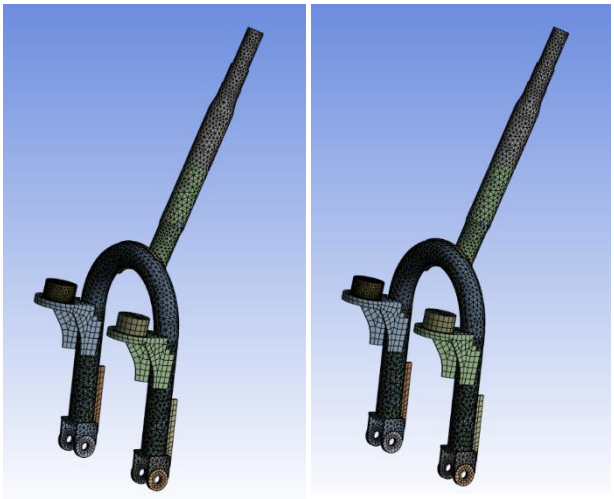


Figure 7.2. a) mesh of element size 10 mm b) 8mm

7.3. L-shape seat mount

7.3.1. 3D model

The 3D l-shape seat mount was designed in SEIMENS NX. Material is stainless steel (Young's modulus= 212 GPa, Density 7885 kg/m^3, tensile strength 355 MPa)

7.3.2. Finite element analysis

Figure 7.2 shows optimized mesh of SOLID 187 element size 6mm. Two ends are fixed to cabin wall and driver seat bottom with 150 kg distributed load and displacement free in x,y,z directions total deformation obtained is 4.3 mm and maximum stress is 275.8 MPa.

7.4. Spring design

To design the spring and determine the stress developed in the spring consider a helical spring subjected to an axial load P.

Let, D = Mean diameter of the spring coil (mm), Do = Outside diameter of the spring coil (mm), Di = Inside diameter of the spring coil (mm), d = Diameter of the spring wire (mm), N = Number of active coils, Nt = Total number of coils, G = Modulus of rigidity for the spring material, P = Axial load on the spring (N or Kg), τ = Max. Shear stress induced in the wire (N/mm2), C = Spring index = D/d, p = Pitch (mm), δ = Deflection of the spring, as a result of an axial load P, L = Free Length (mm), K = Wahl factor, k = Stiffness of the spring (N/mm).

The equations required for the design of the spring are as follows;

a) Spring Index (C) = D/d……..**[equation 7.1]**

b) Wahl's Stress factor (K) = $\frac{(4C-1)}{(4c-4)} + \frac{0.615}{4}$**……[equation 7.2]**

c) Stiffness , $\boldsymbol{k = \frac{Gd^4}{8nD^3}}$ N/mm ……**[equation 7.3]**

d) Shear Stress (τ) = $\boldsymbol{\tau = \frac{K \cdot 8 \cdot P \cdot C}{\pi \cdot d^2}}$**……[equation 7.4]**

e) Solid Length = $\mathbf{N_t \times d}$**……[equation 7.5]**

f) Deflection (δ) = **P/k** mm

Calculations

a) Coil diameter, $d = \frac{D}{C} = \frac{78}{8} = 9.75\,\text{mm}$

b) Number of active coils $n = \frac{79,300 \times (9.75)^4}{8 \times 24.8 \times (78)^3}$

The spring design using the optimized design constraints resulted in a spring of diameter 78 mm, wire diameter 9.75 mm and stiffness 24.5 N/mm

Enhancing one aspect can often negatively impact the other. Choosing the right damping co-efficient and spring stiffness plays a vital role in the suspension performance. General equation of motion is expressed as shown in (7.6):

$[\boldsymbol{M}]\{\boldsymbol{z}\}^{\cdot\cdot} + [\boldsymbol{C}]\{\boldsymbol{z}^{\cdot}\} + [\boldsymbol{K}]\{\boldsymbol{z}\}\ \{\boldsymbol{f(t)}\}$ (7.6)

$\boldsymbol{M_s z_1^{\cdot\cdot}} + \boldsymbol{C_s}(\boldsymbol{z_2^{\cdot}} - \boldsymbol{z_1^{\cdot}}) + \boldsymbol{K_s}(\boldsymbol{z_2} - \boldsymbol{z_1}) + \boldsymbol{K_t z_2} - \boldsymbol{K_t Y_0} = \boldsymbol{0}$(4.7)

$\boldsymbol{M_u z_2^{\cdot\cdot}} + \boldsymbol{C_s}(\boldsymbol{z_1^{\cdot}} - \boldsymbol{z_2^{\cdot}}) + \boldsymbol{K_s}(\boldsymbol{z_1} - \boldsymbol{z_2}) = \boldsymbol{0}$ (7.8)

It is postulated that all dampers and springs behave linearly and equation (4.1) is valid for system of motion. The forcing function $\{f(t)\}$ depends on the tire stiffness value and road disturbance Y_0.

Road Profiles

Numerous studies have been conducted to depict the intricate nature of road surface disruptions. However, creating a single model that accurately captures all the diverse road profile variations at different speeds proves to be a complex challenge, However as shown in Figure 4.2 a standard road profile bump of height 0.1 metre and length is 3.7 metre following IRC-99-1988-2016 is modeled and as shown in Figure 4.3 it is also plotted in time domain using MATLAB. The safe of vehicle speed approaching the bump is 6.94 m/s.

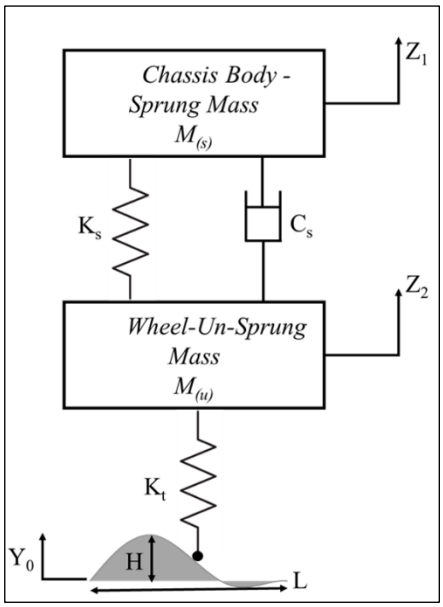


Figure 7. vehicle suspension model over a speed bump

Table 7.3 Essential data for new spring design

New Design Inputs and constraints	Old spring Data	New spring Data
Maximum operating Load	1500 kgs	1500 kgs
Geometrical constraints	L * B * H = 300 * 150 * 90	L * B * H = 300 * 150 * 90
Material	AISI 4454	AISI 4454
Fabrication cost	Same as old design	3800\- INR per unit
Stiffness	Reduction needed	K= 34.792 N/mm

Table 7.4 new design inputs and Old spring data

Essential data for the Spring design	
Parameter	**value**
Axial load, P	3500 N
Spring Diameter, D	86 mm
Coil diameter, d	11.8 mm
Pitch, p	12 mm
Free length, L	255 mm
Material Shear Modulus, G	79 GPa
Active coils	12
Total coils	14

7.5. fabrication and replacing new components on vehicle

7.5.1. U-fork steering column



Figure7.3 a) U-fork prototype with suspension spring b)Suspension housing c)Mounting bracket to swing arm d) Replacing of new components at Greaves testing area

The fabrication U-fork and steering column was done at Greaves Electric Mobility workshop. The work order was given and the prototype was received after a few days and new spring was supplied by the authorised vendor. The L-shape seat mount was fabricated at the same place and shipped to the Greaves R&D office in Bengaluru. Figure 7.6 shows replacing of new components on vehicle. Figure 7.7. shows old design where seat mount was directly attached to chassis member. Figure 7.8 shows new seat mount design attached to cabin wall.

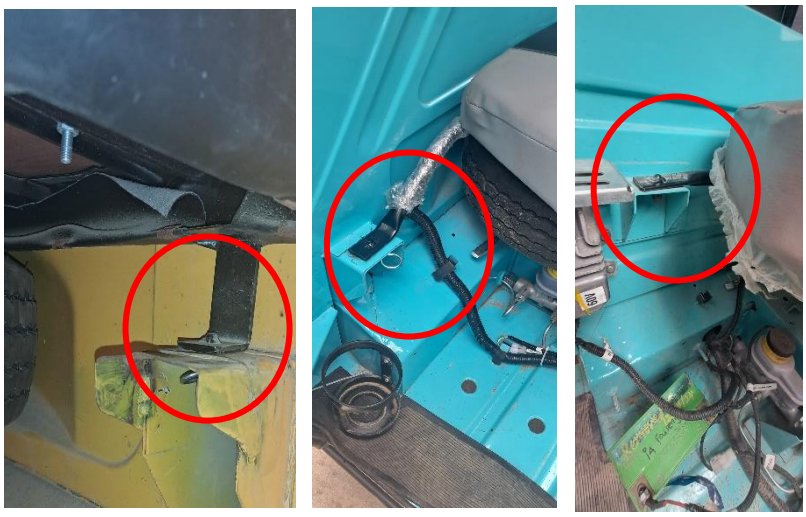


Figure 7.4 Old seat mount position on chassis member

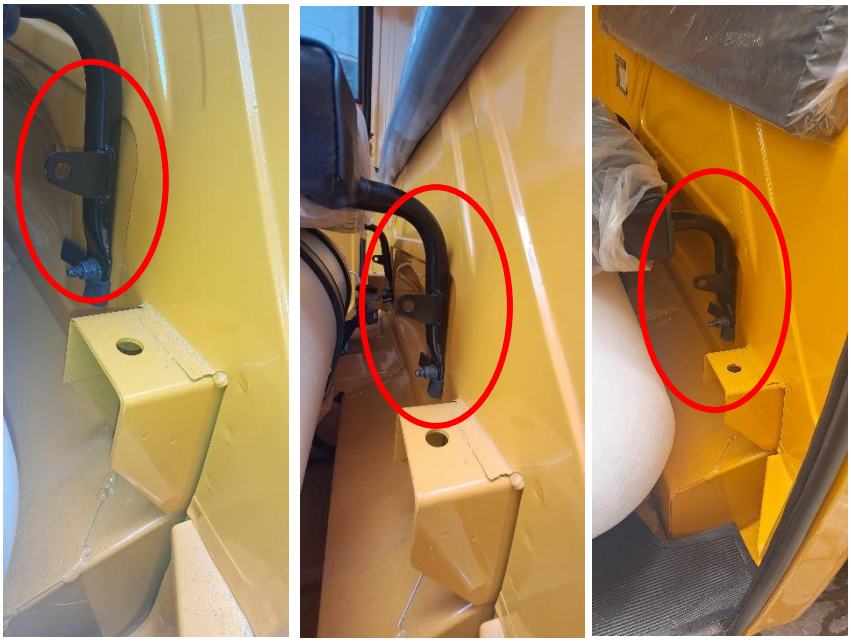


Figure 7.5 New seat mount position fitted to cabin wall

CHAPTER 8

TESTING AND VALIDATION

8.1. Testing of spring

The existing suspension spring weighing 2.90 Kg is heavy hence needed to be optimized and a lighter design of the spring is needed. The existing suspension spring having wire diameter 11.6 mm, with outside and inside diameter 86 mm and 74 mm respectively, total number of coils 14, number of active coils 12 and free length 255mm. New designs have same data as of existing spring only total numbers of coils changed so active coils are also changed. The three designs having 12, 13.5 and12 total coils having 2 coils inactive in each case respectively. By using above equations different values can be found for new developed springs.



Figure 8.1 UTM facility at RVCE Bengaluru

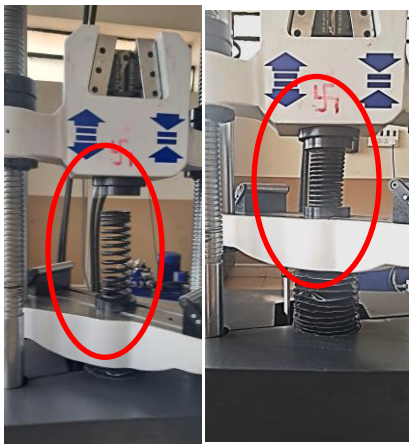


Figure 8.2 a) new spring in no load condition b) fully compressed condition

8.2. RLDA and Whole-Body vibrations test

The seat mount and steering column with new spring was integrated on the vehicle to carry out RLDA and WBV test as per the same DOE used for investigation. Table 8.1 shows the responses of VDV and percentage decrease from original values.

Table 8.1 PSD and VDV values of RLDA for new design

Run	PSD ($m^2/s^3/Hz$)	VDV (Lower Back) (m/s1.75)	VDV (Hand-Arm) (m/s1.75)
1	0.017	4.36	2.62
2	0.018	4.48	2.67
3	0.019	4.50	2.70
4	0.022	5.13	2.99
5	0.023	5.35	3.14
6	0.023	5.15	3.01
7	0.026	5.77	3.30
8	0.028	6.18	3.55
9	0.031	6.50	3.73
10	0.032	6.84	4.28
11	0.033	6.95	4.35
12	0.036	7.33	4.57
13	0.036	7.43	4.54
14	0.038	7.69	4.71
15	0.041	8.13	4.96
16	0.042	8.40	5.04
17	0.044	8.84	5.32
18	0.043	8.44	5.06
19	0.047	9.26	5.89
20	0.050	9.82	6.26
21	0.049	9.53	6.05
22	0.049	9.64	6.02
23	0.052	10.11	6.33
24	0.052	9.91	6.18
25	0.055	10.68	6.57
26	0.057	10.96	6.76
27	0.061	11.59	7.12

CHAPTER 9

RESULTS AND DISCUSSIONS

9.1. Optimization results

Interaction plots for factors and PSD values is shown in figure 9.1. and figure 9.2 shows residual plots. The significance of road roughness effect on PSD is very high indicated by F-ratio value of 172654.

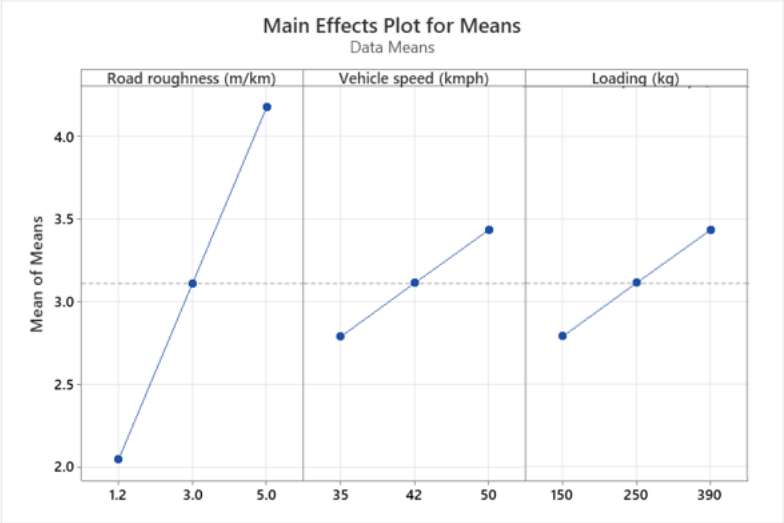


Figure 9.1 shows interaction plot between PSD and factors

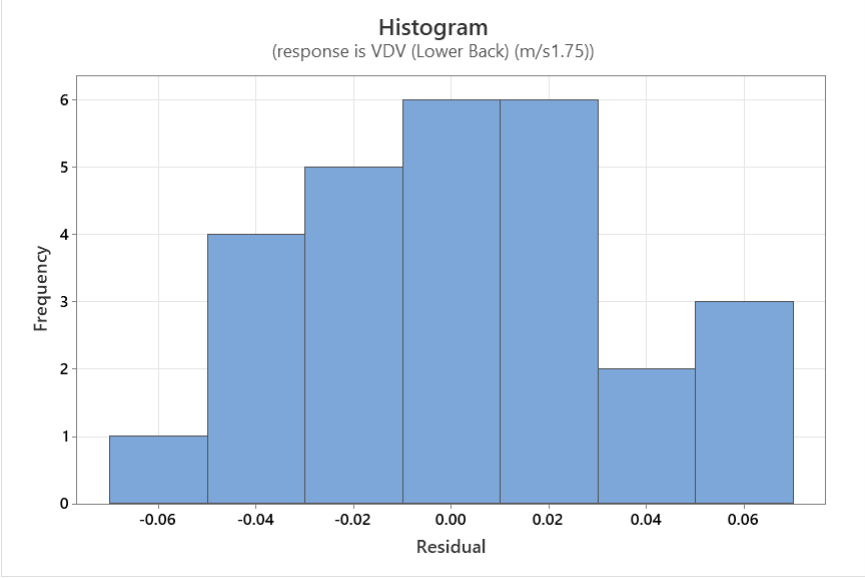


Figure 9.2 residual plots for actual and predicted values

9.2. Spring test Results

The new spring with reduced stiffness contributed for increased damping and thus helped to reduce transmissibility. The stiffness has been reduced to 24.5 N/mm from old value of 34.56 N/mm.

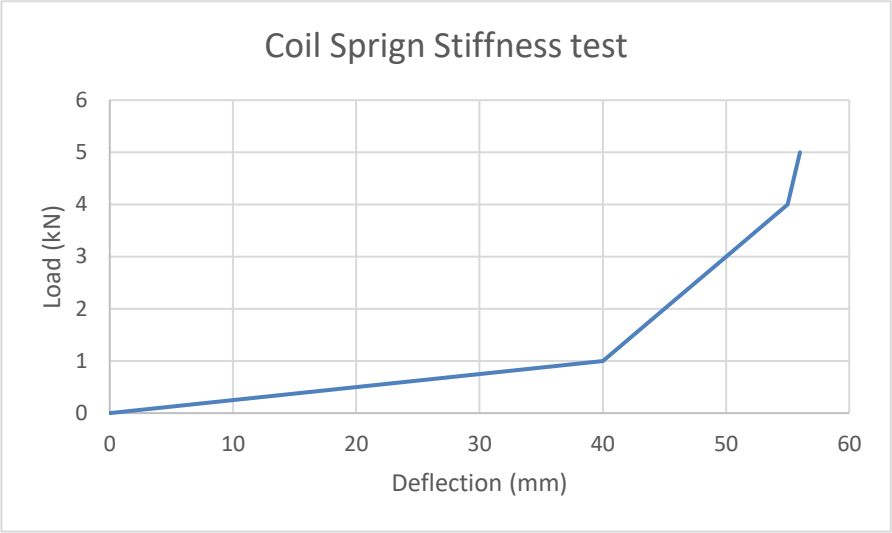


Figure 9.3 Load v/s deflection plot for new spring indication reduced stiffness

9.3. FEA results of new steering column

Modes 1 to 6 and first 6 natural frequencies are obtained. Figure 9.4 shows mode shapes and table 9.1 shows natural frequencies and damping ratios.

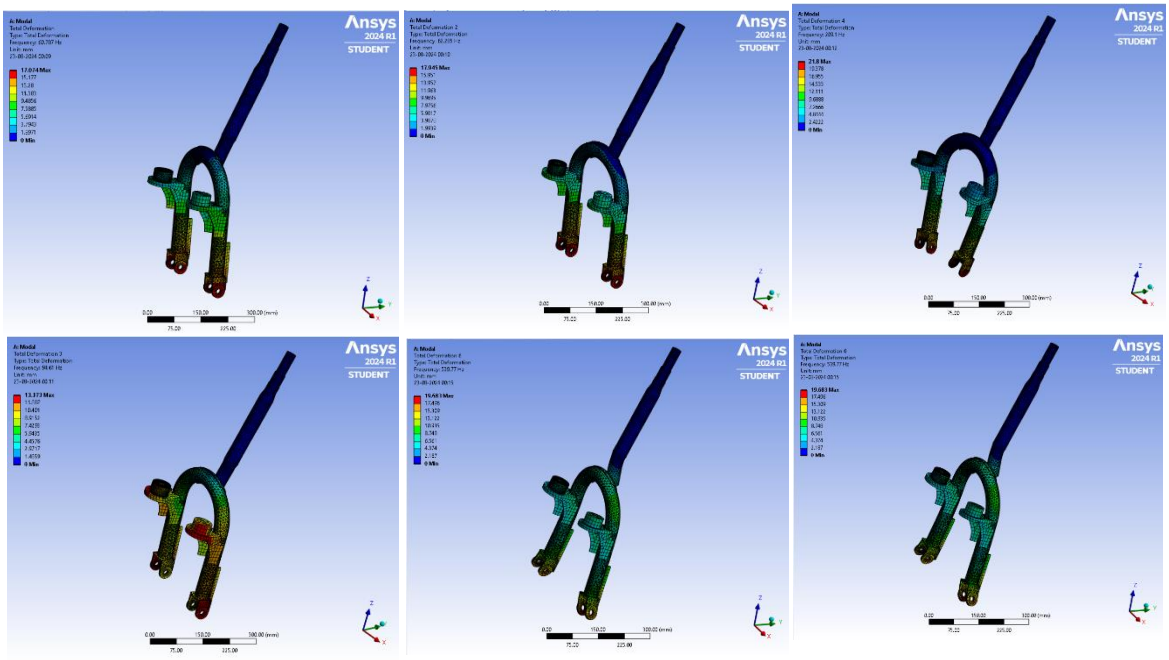


Figure 9.4 modes 1 to 6 of U-fork steering column

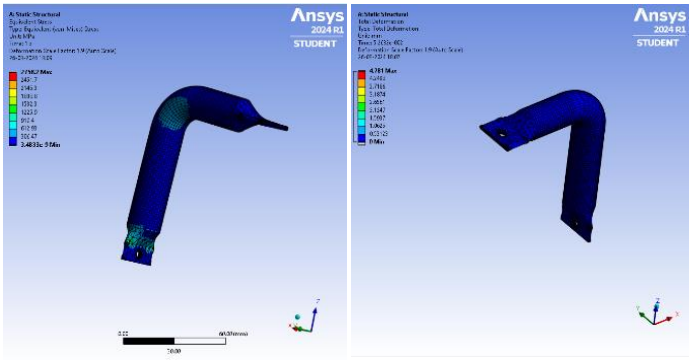


Figure 9.5 a) Max stress of 275.8 MPA b) total deformation of 4.3 mm

Table 9.1 natural frequencies 1 to 6 of new steering column and damping ratios

Mode number	Natural frequency (Hz)	Damping ratios
1	11.3	0.03
2	32.2	0.034
3	57.5	0.089
4	69.3	0.056
5	72.43	0.06
6	79.51	0.095

9.2. RLDA plots

The PSD values obtained during RLDA after implementation of new design showed reduction of around 15% in each response leading to decrease of 13 % Vibration dose values since VDV is a derivative of RMS accelerations, as per the equations provided in previous chapter. Figure 9.6 to figure 9.12 shows comparison plot of PSD plot for old and new design incorporated in cabin at significant ride conditions.

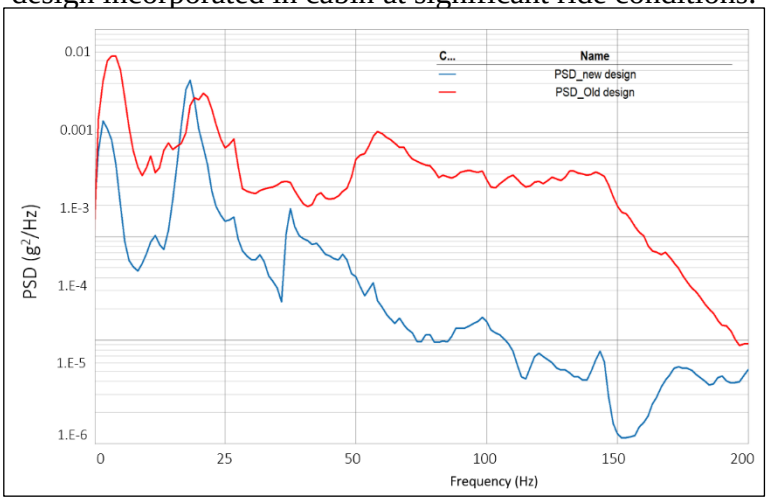


Figure 9.6 PSD at steering- handle bar interface at ride conditions 1-9

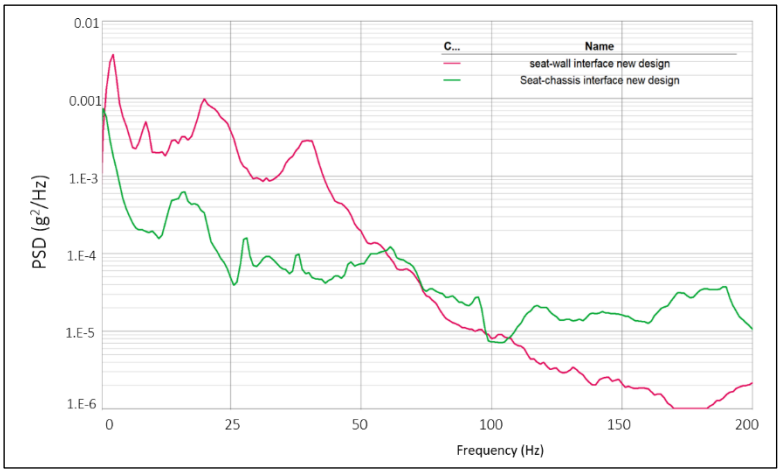


Figure 9.7 PSD at steering- handle bar interface at ride conditions 10-18

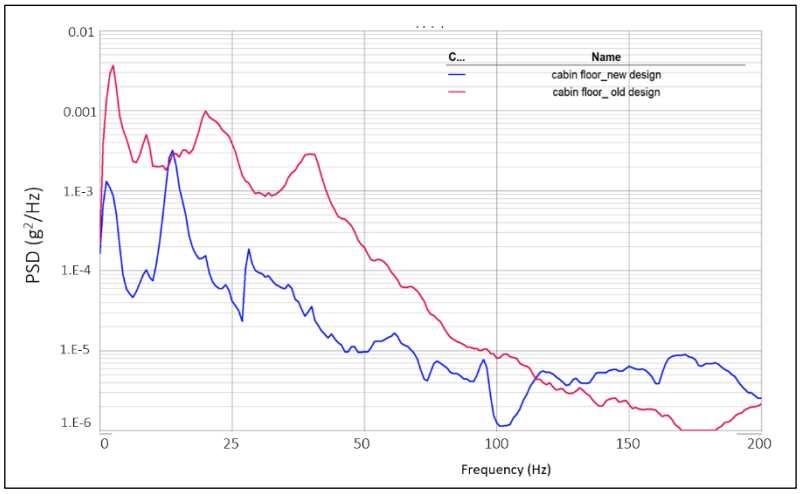


Figure 9.8 PSD at steering- handle bar interface at ride conditions 10-18

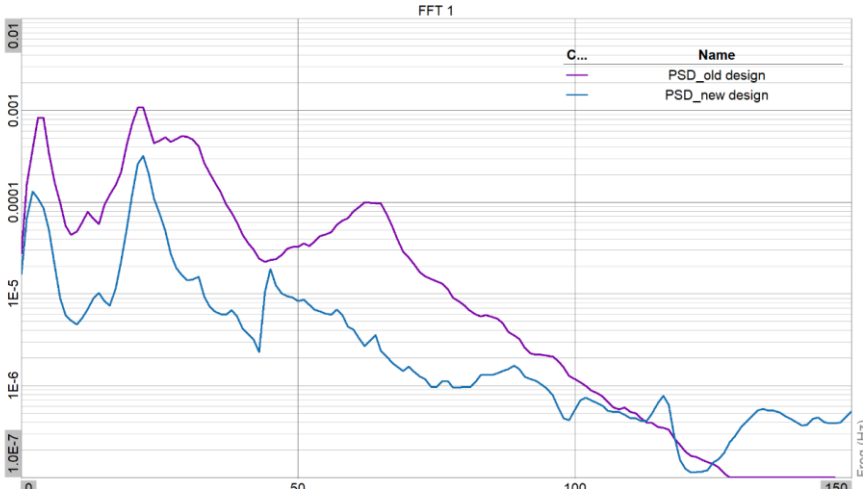


Figure 9.9 Comparison Overall PSD plot for all ride conditions

CHAPTER 10

CONCLUSIONS AND FUTURE SCOPE

Significant amount of work has been done and following conclusions are drawn. Furthermore, a lot of scope is there to carry out advanced testing using test rigs and use of ML algorithms for optimization.

10.1. Conclusions

In order to address the issue of vibration transmissibility onto driver's body from vehicle and road loads, vibration characteristics of vehicle are analysed through modal test and RLDA.

1. The first two modal frequencies of new steering column are 83 Hz and 125 Hz which are well above vehicle's frequency indicating the reduced risk of resonance during ride.
2. RLDA values crossed 0.02 g^2/Hz PSD and 11 $m/s^{1.75}$ VDV. This demands modification in seat mount and steering column.
3. Optimization of factors indicated road roughness index of range of 2 m/km to 3 m/km vehicle speed of 43 kmph and load of 315 kg as optimized values for vehicle ride condition.
4. The Road load test of vehicle with new steering column, suspension spring and seat mount showed reduction in PSD around 12 % and VDV around 13 %.

10.2. Future scope

- The optimized factors showed the requirement of mass addition to steering column by 8 kg and reduction of stiffness of spring by 6%. Hence, future can include design of adaptive shock absorber with varying stiffness according to road roughness.
- Transmissibility of vibration from motor mounting was 92% and from steering column was 72%. Therefore, the motor mountings position and steering column trail position can be varied in vehicle design in next phase to altogether arrest vibration at source instead of damping them.

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